ISOENERGETIC KAM, CONFINEMENT AND THE RESTRICTED THREE-BODY PROBLEM: BLUEPRINT

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Abstract. A proof plan for the four frozen statements of track I: the isoenergetic a-posteriori KAM theorem (I-G1), all-time action confinement for two degrees of freedom (I-HEAD), the Kepler bordered determinant (I-KEPLER), and the near-circular region of the restricted circular planar three-body problem (I-R3BP). Results of the blueprint of track K are cited by their labels [K:`bp:p1:...`] and [K:`bp:p2:...`]. Every result is labelled and cites its pinned source.

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1. Conventions, definitions and the frozen statements

This part fixes notation and restates, in mathematical form, the four frozen theorems of track I and the frozen definitions they use: the fifteen definitions of `KAM/Frozen/Iso/Defs/` (owner approval D20, 27 October 2026) and the definitions of `KAM/Frozen/Defs/` of track K. The Lean declarations are authoritative; the statements below are faithful transcriptions, with the quantifier order and the

order in which constants are chosen exactly as in Lean. Results of track K's blueprint are cited as [K:bp:p1:...] and [K:bp:p2:...]; they are not re-proved here.

1.1. Conventions.

- Ambient spaces are $\mathbb{R}^m = (\texttt{Fin}\ m \to \mathbb{R})$ and $\mathbb{C}^m = (\texttt{Fin}\ m \to \mathbb{C})$. Angles are $\theta \in \mathbb{C}^n$, actions $I \in \mathbb{C}^n$, phase points $z = (\theta, I) \in \mathbb{C}^n \times \mathbb{C}^n$. Functions of θ are 1-periodic, as in Part 0 of track K ([K:bp:p0:conventions]).
- *Norms.* $|v| = \|v\| = \max_i |v_i|$ on $\mathbb{R}^m$ and $\mathbb{C}^m$; on a product the maximum of the two component norms. This is the norm of `Metric.ball`, of `dist` in `actionNbhd`, of $\|DH\|$ (operator norm of a $\mathbb{C}$-linear map, `fderiv`), and of $\|D^2H\|$ (norm of the bilinear map `iteratedFDeriv`). A row-sum bound $\sum_j |M_{ij}| \le c$ for every i is written $|M| \le c$. The norm of a complex number is its modulus.
- The strip $T_\rho = \{\theta \in \mathbb{C}^n : |\operatorname{Im}\theta_i| < \rho\ \forall i\}$ is open; for $\rho > 0$ it contains $\mathbb{R}^n$, and it is invariant under $\mathbb{Z}^n$ and $\mathbb{R}^n$ translations. The set `Set.pi univ (Icc 0 1)` $= [0,1]^n$ carries Lebesgue measure, of total mass one; $\langle u \rangle = \int_{[0,1]^n} u(\theta)\, d\theta$ for a function evaluated at real angles. Averages of vectors or matrices are entrywise.
- A map is *holomorphic on an open set* U if it is `DifferentiableOn` $\mathbb{C}$ there. On an open set all derivatives `fderiv`, `iteratedFDeriv` used below are genuine (Lean totality, Remark 1.13).
- Real and complex points. A real vector $x \in \mathbb{R}^m$ is identified with $(x_i) \in \mathbb{C}^m$; "$(I : \mathbb{C}) \in S$" for $S \subset \mathbb{C}^n$ means this identification lies in S. The real phase points of $D \subset \mathbb{C}^n \times \mathbb{C}^n$ are the $(\theta, I) \in \mathbb{R}^n \times \mathbb{R}^n$ whose image lies in D.
- *Two phase-space conventions.* In I-G1 and I-HEAD, a phase point is (θ, I) (angles first) and $X_H = (\partial_I H, -\partial_\theta H)$. In the restricted three-body problem a phase point is $z = (q, p)$ with position $q \in \mathbb{R}^2$ and canonical momentum $p \in \mathbb{R}^2$ (position first, as in the Lean carrier $(\texttt{Fin}\ 2 \to \mathbb{R}) \times (\texttt{Fin}\ 2 \to \mathbb{R})$); there `radius` is the Euclidean length, not the max norm. The norm $\|\cdot\|$ of I-HEAD is the max norm; the bounds in the conclusion of I-R3BP use rad.

1.2. Frozen definitions of track K used by track I.

Definition 1.1 (Strip, action neighbourhood)**.** $T_\rho = \{\theta \in \mathbb{C}^n : \forall i,\ |\operatorname{Im}\theta_i| < \rho\}$ (`complexStrip` n ρ). For $B \subset \mathbb{R}^n$ and $r \in \mathbb{R}$, $B_r = \{I \in \mathbb{C}^n : \exists J \in B,\ \|I - J\| < r\}$ (`actionNbhd`); it is open. Thus for real I: $I \in B_r$ iff $\exists J \in B$ with $\|I - J\| < r$.
Frozen Lean declaration: `KAM.complexStrip, KAM.actionNbhd`
Source: HL19 §2.1; see [K:bp:p0:domains].

Definition 1.2 (Diophantine vectors)**.** $\omega \in \mathbb{R}^n$ is (γ, τ)-Diophantine if for every $k \in \mathbb{Z}^n \setminus \{0\}$ $\gamma / \big(\sum_i |k_i|\big)^\tau \le |k \cdot \omega|$ (real power).
Frozen Lean declaration: `KAM.IsDiophantine`
Source: HL19 (3.1), p. 14; see [K:bp:p0:diophantine].

Definition 1.3 (Hamiltonian vector fields)**.** For $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ and a real phase point z: $X_H(z) = \big(i \mapsto DH(z)(0, e_i),\ i \mapsto -DH(z)(e_i, 0)\big)$, with D the real Fréchet derivative. For $H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ the same formula with the complex derivative gives $X_H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}^n \times \mathbb{C}^n$. Thus $X_H = (\partial_I H, -\partial_\theta H)$.
Frozen Lean declaration: `KAM.hamiltonianVectorField, KAM.hamiltonianVectorField`$\mathbb{C}$
Source: HL19 Rem. 2.1, p. 6.

Definition 1.4 (Real Hamiltonian, frequency map)**.** For $h : \mathbb{C}^n \to \mathbb{C}$ and $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ and a real phase point (θ, I), $\mathcal{H}_{h,f}(\theta, I) = \operatorname{Re}\big(h(I) + f(\theta, I)\big)$ (`realHamiltonian`). The frequency map is $\nabla h(I)_i = D\big(J \mapsto \operatorname{Re} h(J)\big)(I)\, e_i$ for real I (`frequencyMap`), a real function $\mathbb{R}^n \to \mathbb{R}^n$.
Frozen Lean declaration: `KAM.realHamiltonian, KAM.frequencyMap`
Source: see [K:bp:p0:nearlyintegrable].

Definition 1.5 (Torus data on strips)**.** Let $K : \mathbb{C}^n \to \mathbb{C}^n \times \mathbb{C}^n$, $H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$, $\omega \in \mathbb{R}^n$, $\theta \in \mathbb{C}^n$.

- K is *primary on* T_ρ if $K(\theta+k) = K(\theta) + (k,0)$ for $\theta \in T_\rho$, $k \in \mathbb{Z}^n$ (`IsPrimaryOnStrip`); *real on reals* if every $K(\theta)$, $\theta \in \mathbb{R}^n$, is a real phase point (`IsRealOnReals`).
- Bilinear dot product $u \cdot v = \sum_i (u_{1,i}v_{1,i} + u_{2,i}v_{2,i})$ on $\mathbb{C}^n \times \mathbb{C}^n$ (`dotℂ`), $\Omega_0(a,b) = (-b,a)$ (`omegaZero`).
- Invariance error $E_{H,K,\omega}(\theta) = X_H(K(\theta)) - DK(\theta)\omega$ (`invarianceError`).
- Gram matrix $G_K(\theta)_{ij} = DK(\theta)e_i \cdot DK(\theta)e_j$ (`gramMatrix`), a complex $n \times n$ matrix.
- Normal frame $N_K(\theta)_{\cdot j} = \sum_k (G_K^{-1})_{kj}\, \Omega_0\big(DK(\theta)e_k\big) \in \mathbb{C}^n \times \mathbb{C}^n$ (`normalFrame`), with the matrix inverse of Lean.
- Torsion $T_{H,K,\omega}(\theta)_{ij} = N_{\cdot i}(\theta) \cdot \Omega_0\big(DX_H(K(\theta))N_{\cdot j}(\theta) - D_\theta(N_{\cdot j})(\theta)\,\omega\big)$ (`torsion`); $N_{\cdot j}$ is regarded as a function of θ.

Frozen Lean declaration: `KAM.IsPrimaryOnStrip, KAM.IsRealOnReals, KAM.dotℂ, KAM.omegaZero, KAM.invarianceError, KAM.gramMatrix, KAM.normalFrame, KAM.torsion`
Source: HL19 (2.2), (3.2)–(3.9), pp. 7, 16–17; see [K:`bp:p0:stripdata`].

1.3. **Frozen definitions of track I.**

Definition 1.6 (Energy average and extended torsion)**.** For $H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$, K as above, $\omega \in \mathbb{R}^n$, $\theta \in \mathbb{C}^n$, with index set $\{1,\dots,n\} \sqcup \{*\}$:

$$\bar{E}_H(K) = \int_{[0,1]^n} H\big(K(\theta)\big)\, d\theta \in \mathbb{C}, \qquad \mathsf{T}_H(\theta) = \begin{pmatrix} T_{H,K,\omega}(\theta) & \omega \\ \big(DH(K(\theta))N_{\cdot b}(\theta)\big)_b & 0 \end{pmatrix},$$

the $(n+1)\times(n+1)$ complex matrix whose upper-left block is the torsion, last column $(\omega, 0)$ (real entries, viewed in $\mathbb{C}$), and last row $b \mapsto DH(K(\theta))\big(N_{\cdot b}(\theta)\big)$ (complex derivative of H applied to the normal frame), last entry 0. The averaged extended torsion is $\langle \mathsf{T}_H \rangle = \big(\int_{[0,1]^n} \mathsf{T}_H(\theta)_{ij}\, d\theta\big)_{ij}$, the integral taken over real angles.
Frozen Lean declaration: `KAM.Iso.energyAverage, KAM.Iso.extendedTorsion, KAM.Iso.extendedTorsionAverage`
Source: HL19 (3.13)–(3.14), p. 19, with $c = H$, $d = n$ (Case III), $\hat{\omega} = \omega$; the last row is $DH(K)N$ even off invariance, not $\omega^\top$ (HL19 (3.17) is the invariant case, Rem. 3.7, p. 19).

Definition 1.7 (Bordered Hessian)**.** For $h : \mathbb{C}^n \to \mathbb{C}$ and real I, the real $(n+1)\times(n+1)$ matrix

$$\mathsf{B}_h(I) = \begin{pmatrix} \big(\partial_b(\nabla h)_a(I)\big)_{a,b} & \nabla h(I) \\ \nabla h(I)^\top & 0 \end{pmatrix},$$

where $\partial_b(\nabla h)_a(I) = D(\nabla h)(I)e_b$ read at a, the real derivative of the frequency map (`borderedHessian`). No symmetry is assumed in the definition.
Frozen Lean declaration: `KAM.Iso.borderedHessian`
Source: Chi05 p. 17 (display after (3.43)); HL19 Rem. 3.7, p. 19.

Definition 1.8 (Hamiltonian solutions with action in B)**.** For $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, $B \subset \mathbb{R}^n$, $J \subset \mathbb{R}$ and $z : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$, write $\mathrm{Sol}(H, B, J, z)$ if for all $t \in J$: $z(t)_2 \in B$, H is real differentiable at $z(t)$, and z is differentiable at t with $z'(t) = X_H(z(t))$ (`HasDerivAt`).
Frozen Lean declaration: `KAM.Iso.IsHamiltonianSolutionOn`
Source: ours (D20); the domain B and the differentiability of H are part of the solution concept.

Definition 1.9 (Planar Euclidean length, the primaries, collisions)**.** For $q \in \mathbb{R}^2$: $\rho_2(q)^2 = q_0^2 + q_1^2$ (`radiusSq`) and $\mathrm{rad}(q) = \sqrt{q_0^2 + q_1^2}$ (`radius`; the radicand is nonnegative). For $\mu \in \mathbb{R}$: $a_1(\mu) = (-\mu, 0)$, $a_2(\mu) = (1-\mu, 0)$ ($\mathtt{primary}_1$, $\mathtt{primary}_2$). A phase point $z = (q,p) \in \mathbb{R}^2 \times \mathbb{R}^2$ is *collision free* (`CollisionFree` μ) if $\mathrm{rad}(q - a_1(\mu)) > 0$ and $\mathrm{rad}(q - a_2(\mu)) > 0$.
Frozen Lean declaration: `KAM.Iso.radiusSq, KAM.Iso.radius, KAM.Iso.primary₁, KAM.Iso.primary₂, KAM.Iso.CollisionFree`
Source: Chi05 (3.34), (3.37), p. 12–13 (barycentric rotating frame, unit separation, total mass one); ours.

Definition 1.10 (Restricted Hamiltonian and field)**.** On collision-free points $z = (q, p)$, with $r_j = \mathrm{rad}(q - a_j(\mu))$:

$$\mathcal{H}^{\mathrm{R3}}_\mu(q,p) = \tfrac{1}{2}|p|_2^2 + q_1p_0 - q_0p_1 - \frac{1-\mu}{r_1} - \frac{\mu}{r_2}, \qquad |p|_2^2 := p_0^2 + p_1^2,$$

$$V_\mu(q,p) = \Big((p_0 + q_1,\ p_1 - q_0),\ (p_1, -p_0) - \frac{1-\mu}{r_1^3}(q-a_1) - \frac{\mu}{r_2^3}(q-a_2)\Big).$$

Thus $V_\mu = (\partial_p \mathcal{H}^{\mathrm{R3}}_\mu, -\partial_q \mathcal{H}^{\mathrm{R3}}_\mu)$ on the open collision-free set. A function $z : \mathbb{R} \to \mathbb{R}^2 \times \mathbb{R}^2$ is a *restricted solution on* J, $\mathrm{RSol}_\mu(J, z)$, if for every $t \in J$ the point $z(t)$ is collision free and z is differentiable at t with $z'(t) = V_\mu(z(t))$.

Frozen Lean declaration: `KAM.Iso.restrictedHamiltonian, KAM.Iso.restrictedVectorField, KAM.Iso.IsRestrictedSolutionOn`

Source: Chi05 (1.2), p. 2 and (3.34)–(3.39), pp. 12–14, in barycentric rotating coordinates with canonical momentum (not rotating velocity); the equations V_μ are the Hamilton equations of $\mathcal{H}^{\mathrm{R3}}_\mu$ (the check is part of the proofs, see PLAN.md).

Definition 1.11 (Near-circular region)**.** For $a_-, a_+, e_-, e_+ \in \mathbb{R}$, $\mathcal{N}(a_-, a_+, e_-, e_+) \subset \mathbb{R}^2 \times \mathbb{R}^2$ is the set of (q, p) for which there exist $r, a, e \in \mathbb{R}$ with

$$r > 0,\ r^2 = q_0^2 + q_1^2, \quad a > 0, \quad 0 < e < 1, \quad a_- < a < a_+, \quad e_- < e < e_+,$$

$$\tfrac{1}{2}|p|_2^2 - \frac{1}{r} = -\frac{1}{2a}, \qquad q_0p_1 - q_1p_0 > 0, \qquad (q_0p_1 - q_1p_0)^2 = a(1-e^2).$$

(`NearCircular`.) The three numbers r, a, e are determined by (q, p), so no choice is involved.

Frozen Lean declaration: `KAM.Iso.NearCircular`

Source: Chi05 (2.24)–(2.26), p. 9: $E = -1/(2a)$, $\Gamma^2 = a(1-e^2)$ in the units $\mu = M = 1$; referred to $\mu = 0$ (primary at the origin).

Definition 1.12 (The Kepler bordered matrix)**.** For $\kappa, \Omega \in \mathbb{R}$ and $L > 0$,

$$\mathsf{K}_{\kappa,\Omega}(L) = \begin{pmatrix} -3\kappa/L^4 & 0 & \kappa/L^3 \\ 0 & 0 & -\Omega \\ \kappa/L^3 & -\Omega & 0 \end{pmatrix} \in \mathbb{R}^{3\times 3},$$

the bordered Hessian $\left(\begin{smallmatrix} D^2h & \nabla h \\ \nabla h^\top & 0 \end{smallmatrix}\right)$ of $h(L, G) = -\kappa/(2L^2) - \Omega G$ (`keplerBorder`, L in the subtype $\{L : \mathbb{R} \mid L > 0\}$).

Frozen Lean declaration: `KAM.Iso.keplerBorder`

Source: Chi05 p. 17 ($\kappa = \Omega = 1$).

1.4. **Lean totality.**

Remark 1.13 (Lean totality). The frozen definitions use `fderiv` (zero where a function is not differentiable), the matrix inverse (zero for a singular matrix), the Bochner integral (zero for a non-integrable function), division, `Real.sqrt` (zero for a negative argument) and real powers. The statements guard them as follows; each proof must establish the guarded facts before using the defining formulas.

- (T1) *Derivatives of H in I-G1.* D is open, H is holomorphic on D, so DH, X_H, DX_H and D^2H are genuine derivatives at every point of D; K is holomorphic on the open strip T_ρ and $B_\mu(K(\theta)) \subset D$, so $H \circ K$, $X_H \circ K$ and DK are genuine on T_ρ.
- (T2) *Inverses in I-G1.* The Gram matrix inverse is used only under the hypothesis $\det G_K(\theta) \in \mathbb{C}^\times$ on T_ρ, and the inverse of the averaged extended torsion only under $\det\langle \mathsf{T}_H \rangle \in \mathbb{C}^\times$. Then N_K is genuine and holomorphic on T_ρ (by the holomorphic inverse, [K:`bp:p1:holomorphic-inverse`]), so $DN_{\cdot j}$ and T are genuine. Gram invertibility must be proved before using a normal frame or torsion for K_∞; no such fact is needed to interpret the frozen final invariance error or energy equality, which use only DK_∞ and X_H.

(T3) *Integrals in I-G1.* The cube $[0,1]^n \subset \mathbb{R}^n \subset T_\rho$ is compact, $\rho > 0$, and the integrands $H \circ K$ and the entries of T_H are continuous on it by (T1)–(T2), hence integrable ([K:`bp:p1:ext-integral`]). Integrability is not a hypothesis; it must be proved.

(T4) *Powers and division in I-G1.* $\gamma > 0$ and $\rho > 0$, so γ^4, γ^2, $\rho^{4\tau}$, $\rho^{2\tau}$ are positive and the quotients are genuine; $\gamma/(\sum|k_i|)^\tau$ has a positive base since $k \neq 0$.

(T5) *I-HEAD.* h is holomorphic on the open set B_{r_0}, so ∇h and $D(\nabla h)$ are genuine at real points of B_{r_0} (an open subset of $\mathbb{R}^2$) and B_h is genuine. The solution predicate Sol includes differentiability of $\mathcal{H}_{h,f}$ at each visited point, which also guards $X_{\mathcal{H}}$; it holds for $z(t)_2 \in B \subset B_{r_0}$ and real θ because f is holomorphic on $T_{s_0} \times B_{r_0}$.

(T6) *I-KEPLER, I-R3BP.* $L > 0$ guards every denominator of $\mathsf{K}_{\kappa,\Omega}(L)$. Every inverse distance in $\mathcal{H}_\mu^{\mathrm{R3}}$, V_μ is guarded by collision freedom, and the radicand of `radius` is a sum of squares. In $\mathcal{N}$, $r > 0$ and $a > 0$ precede the energy equation.

1.5. **The frozen theorems.**

Theorem 1.14 (I-G1: isoenergetic a-posteriori KAM)**.** *Let $n \geq 2$ and $\tau \in \mathbb{R}$ with $\tau \geq n-1$; let $\rho, \rho_\infty \in \mathbb{R}$ with $0 < \rho_\infty < \rho$; let $M, \sigma_K, \sigma_B, \sigma_T, \mu, \bar\gamma, \sigma_\omega, \nu \in \mathbb{R}$ with $M, \sigma_K, \sigma_B, \sigma_T, \mu, \bar\gamma > 0$, $\sigma_\omega > 1$, $\nu > 0$; let $\omega^* \in \mathbb{R}^n$, $\omega^* \neq 0$. Then there exist $C_1, C_2, C_3 > 0$ such that for all $D \subset \mathbb{C}^n \times \mathbb{C}^n$, $H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$, $K : \mathbb{C}^n \to \mathbb{C}^n \times \mathbb{C}^n$, and $s, \gamma, e, E_0 \in \mathbb{R}$ satisfying*

(I1) D is open and $(\theta + k, I) \in D$ whenever $(\theta, I) \in D$, $k \in \mathbb{Z}^n$;
(I2) H is holomorphic on D, and $H(\theta + k, I) = H(\theta, I)$ on D for $k \in \mathbb{Z}^n$;
(I3) for every real phase point (θ, I) lying in D, $\operatorname{Im} H(\theta, I) = 0$;
(I4) for every $z \in D$: $\|DH(z)\| \leq M$, $\|X_H(z)\| \leq M$, $\|D(X_H)(z)\| \leq M$, $\|D^2(X_H)(z)\| \leq M$, $\|D^2 H(z)\| \leq M$;
(I5) K is holomorphic on T_ρ, primary on T_ρ, and real on reals;
(I6) for every $\theta \in T_\rho$: $B_\mu(K(\theta)) \subset D$ (open max-norm ball) and $\|DK(\theta)\| \leq \sigma_K - \mu$;
(I7) for every $\theta \in T_\rho$: $\det G_K(\theta) \neq 0$ and $|G_K(\theta)^{-1}| \leq \sigma_B - \mu$ (maximal row sum);
(I8) $\det\langle \mathsf{T}_H \rangle \neq 0$ with T_H built from $(H, K, s\omega^*)$, and $|\langle \mathsf{T}_H \rangle^{-1}| \leq \sigma_T - \mu$ (maximal row sum);
(I9) $1 < s < \sigma_\omega$, $\nu \leq (s-1)\|\omega^*\|$, $\nu \leq (\sigma_\omega - s)\|\omega^*\|$;
(I10) $0 < \gamma \leq \bar\gamma$, $\omega^* \in \mathcal{D}_{\gamma,\tau}$, $0 \leq e$;
(I11) $\|E_{H,K,s\omega^*}(\theta)\| \leq e$ for every $\theta \in T_\rho$;
(I12) $|\bar E_H(K) - E_0| \leq e$;
(I13) $C_1 e/(\gamma^4 \rho^{4\tau}) < 1$,

there are $K_\infty : \mathbb{C}^n \to \mathbb{C}^n \times \mathbb{C}^n$ and $s_\infty \in \mathbb{R}$ with $1 < s_\infty < \sigma_\omega$ such that

(O1) K_∞ is holomorphic on T_{ρ_∞}, primary on T_{ρ_∞} and real on reals;
(O2) $E_{H,K_\infty,s_\infty\omega^*}(\theta) = 0$ for all $\theta \in T_{\rho_\infty}$;
(O3) $H(K_\infty(\theta)) = E_0$ for all $\theta \in T_{\rho_\infty}$ (pointwise);
(O4) $B_{\mu/2}(K_\infty(\theta)) \subset D$ and $\|DK_\infty(\theta)\| \leq \sigma_K - \mu/2$ for $\theta \in T_{\rho_\infty}$;
(O5) $\|K_\infty(\theta) - K(\theta)\| \leq C_2 e/(\gamma^2 \rho^{2\tau})$ for $\theta \in T_{\rho_\infty}$;
(O6) $\|s_\infty \omega^* - s\omega^*\| \leq C_3 e/(\gamma^2 \rho^{2\tau})$.

Here C_1, C_2, C_3 are chosen after all of $n, \tau, \rho, \rho_\infty, M, \sigma_K, \sigma_B, \sigma_T, \mu, \bar\gamma, \sigma_\omega, \nu, \omega^*$ and before $D, H, K, s, \gamma, e, E_0$; hence they may depend on the former, and on none of the latter.

Frozen Lean declaration: `KAM.Iso.isoenergetic_kam`

Source: HL19 Theorem 3.6 and Rem. 3.7, pp. 18–19, in the canonical case $d = n$, $c = h$; proof pp. 45–53. Deviations (all conservative, D-I-G1-UNIFORM, D-I-ENERGY, notes.md): ray data, $\bar\gamma$ and the margins are fixed first; the strict inequalities of (3.16) are replaced by $\leq$ (candidate erratum: with $E_c = 0$ the strict bounds read $0 < 0$); the energy conclusion is pointwise.

Remark 1.15 (Reading of I-G1). (a) The frequency of the approximate torus is $\omega = s\omega^*$ on the open ray $\Theta = \{s'\omega^* : 1 < s' < \sigma_\omega\}$. The Diophantine condition is imposed on ω^*, hence (by homogeneity of $k \mapsto k \cdot \omega$) $s'\omega^*$ is nonresonant for every $s' > 0$, with constant $\gamma s'$. (b) ν enters only through

(I9) and through the choice of C_1, C_2, C_3; it measures the distance of $s\omega^*$ to the two *endpoints* ω^* and $\sigma_\omega\omega^*$ of the segment Θ, i.e. $\min(s-1, \sigma_\omega - s)|\omega^*| \geq \nu$ (HL19 H_5', $\mathrm{dist}(\omega, \partial\Theta) > 0$, made quantitative, with $\partial\Theta$ the boundary *relative to the line* $\mathbb{R}\omega^*$; for $n \geq 2$ the boundary of Θ in $\mathbb{R}^n$ is all of $\bar{\Theta}$, so the ambient reading would be vacuous). The frozen Lean states the two endpoint inequalities explicitly. (c) Taking $e = 0$ gives the exact-torus case; then (I13) holds for every C_1, and (O5), (O6) force $K_\infty = K$ on T_{ρ_∞} and $s_\infty = s$. (d) The row-sum bound in (I7)–(I8) is the max-norm operator norm on $\mathbb{C}^n$, $\mathbb{C}^{n+1}$ respectively.

Theorem 1.16 (I-HEAD: all-time action confinement for two degrees of freedom)**.** *Let $B \subset \mathbb{R}^2$ be open, bounded and nonempty; $s_0, r_0 > 0$; h holomorphic on B_{r_0} with $\operatorname{Im} h(I) = 0$ for every real I with $I \in B_{r_0}$ and*

$$\det \mathsf{B}_h(I) \neq 0 \qquad \text{for every real } I \in B_{r_0}.$$

Let $B' \subset \mathbb{R}^2$ be open with $\overline{B'}$ compact and $\overline{B'} \subset B$, and let $\eta > 0$. Then there is $\varepsilon_0 > 0$ such that for every $f : \mathbb{C}^2 \times \mathbb{C}^2 \to \mathbb{C}$ and $\varepsilon \in \mathbb{R}$ with

- *f holomorphic on $T_{s_0} \times B_{r_0}$, $f(\theta + k, I) = f(\theta, I)$ there for $k \in \mathbb{Z}^2$, and $\operatorname{Im} f(\theta, I) = 0$ for real θ and real $I \in B_{r_0}$;*
- *$0 \leq \varepsilon < \varepsilon_0$ and $|f(z)| \leq \varepsilon$ on $T_{s_0} \times B_{r_0}$,*

and every $z_0 = (\theta_0, I_0) \in \mathbb{R}^2 \times \mathbb{R}^2$ with $I_0 \in B'$, there is $z : \mathbb{R} \to \mathbb{R}^2 \times \mathbb{R}^2$ with $z(0) = z_0$, $\mathrm{Sol}(\mathcal{H}_{h,f}, B, \mathbb{R}, z)$, $\|z(t)_2 - I_0\| < \eta$ for all $t \in \mathbb{R}$, and: for all $a < 0 < b$ and all $y : \mathbb{R} \to \mathbb{R}^2 \times \mathbb{R}^2$ with $y(0) = z_0$ and $\mathrm{Sol}(\mathcal{H}_{h,f}, B, (a, b), y)$ one has $y(t) = z(t)$ for all $t \in (a, b)$.

Frozen Lean declaration: `KAM.Iso.all_time_confinement`

Source: ours (D-I-CONFINEMENT): assembled from I-G1, separation in an energy cylinder (Arnold-type barriers; cf. CC03 Theorem 1.1, p. 7, a numerical one-energy precedent with a truncated potential, not used) and ODE continuation.

Remark 1.17 (Reading of I-HEAD). (a) The constant ε_0 depends on B, s_0, r_0, h, B', η only (it may not depend on f, ε, z_0). (b) $\mathsf{B}_h(I)$ is the matrix of Definition 1.7 with $n = 2$; its determinant is the isoenergetic nondegeneracy of HL19 H_4' at the torus $\theta \mapsto (\theta, I)$: for $H = h$, $K(\theta) = (\theta, I)$, $\omega = \nabla h(I)$ one computes $N_{\cdot b} = (0, e_b)$, $T = D(\nabla h)(I)$, last row $\nabla h(I)^\top$, so $\mathsf{T}_h = \mathsf{B}_h(I)$ exactly (to be proved in part p1, see PLAN.md). (c) The conclusion contains global existence for all real t and uniqueness among every solution on any interval $(a, b) \ni 0$ that remains in $\{I \in B\}$; in particular it contains uniqueness of z among solutions confined to B. (d) The norm $\|\cdot\|$ is the max norm on $\mathbb{R}^2$.

Theorem 1.18 (I-KEPLER: the classical isoenergetic determinant)**.** *For all $\kappa, \Omega > 0$ and $L > 0$: $\det \mathsf{K}_{\kappa,\Omega}(L) = 3\kappa\Omega^2/L^4$ and $\det \mathsf{K}_{\kappa,\Omega}(L) > 0$.*

Frozen Lean declaration: `KAM.Iso.kepler_bordered_det`

Source: Chi05 p. 17 ($\kappa = \Omega = 1$: $3/L^4$); general κ, Ω is the same cofactor computation (ours).

Theorem 1.19 (I-R3BP: near-circular region of the restricted three-body problem)**.** *Let $a_-, a_+, e_-, e_+ \in \mathbb{R}$ with $0 < a_- < a_+$, $0 < e_- < e_+ < 1/4$ and $a_+(1 + e_+) < 1$. Then the set $\mathcal{N} = \mathcal{N}(a_-, a_+, e_-, e_+) \subset \mathbb{R}^2 \times \mathbb{R}^2$ is open and nonempty, and there are $\mu_0, d, R \in \mathbb{R}$ with $0 < \mu_0 \leq 1/2$, $d > 0$, $R > 0$ such that for every $\mu \in [0, \mu_0)$ and every $z_0 \in \mathcal{N}$ there is $z : \mathbb{R} \to \mathbb{R}^2 \times \mathbb{R}^2$ with $z(0) = z_0$, $\mathrm{RSol}_\mu(\mathbb{R}, z)$ and for every $t \in \mathbb{R}$*

$$\mathrm{rad}\big(q(t) - a_1(\mu)\big) \geq d, \quad \mathrm{rad}\big(q(t) - a_2(\mu)\big) \geq d,$$
$$\mathrm{rad}(q(t)) \leq R, \quad \mathrm{rad}(p(t)) \leq R, \qquad z(t) = (q(t), p(t)),$$

and: for all $a < 0 < b$ and y with $y(0) = z_0$ and $\mathrm{RSol}_\mu((a, b), y)$, $y(t) = z(t)$ on (a, b). Here μ_0, d, R depend on $a_\pm, e_\pm$ only, not on μ or z_0.

Frozen Lean declaration: `KAM.Iso.restricted_near_circular`

Source: ours (D-I-PHYSICAL), derived from I-HEAD, I-KEPLER and the Delaunay coordinates of Chi05 §2–3, pp. 8–9, 12–17; not the numerical Victoria theorem of CC03.

Remark 1.20 (Reading of I-R3BP). (a) $\mu = 0$ is included (a massless secondary at $(1, 0)$). (b) The hypothesis $a_+(1+e_+) < 1$ says that the apocentre radius $a(1+e)$ of every orbit of the region is below $1-$margin, so the orbits stay inside the orbit of the secondary; $e_+ < 1/4$ and $e_- > 0$ keep the orbits away from the singularities $e = 0$, $e = 1$ of the Delaunay chart. (c) The region is μ-independent and referenced to $\mu = 0$ (the primary at the origin); the primaries are at $a_1(\mu)$, $a_2(\mu)$ in the dynamics. (d) Collision exclusion for all times is part of the conclusion (distances $\geq d$).

2. Extended torsion and the bordered triangular system

This part collects the linear-algebra and first-order geometry of the isoenergetic scheme of HL19 §6 (pp. 45–51), for the frozen objects $\bar{E}_H$, T_H, $\langle \mathsf{T}_H \rangle$ and B_h of Definitions 1.6 and 1.7. It proves five things.

(1) $\mathsf{T}_H = \mathsf{B}_h(I)$ exactly at the trivial torus $\theta \mapsto (\theta, I)$ (Lemma 2.6); this is why the nondegeneracy hypothesis of I-HEAD is the isoenergetic condition of I-G1.
(2) The last row of T_H is $\omega^\top - (\eta^L)^\top$ exactly, hence $\omega^\top$ on an invariant torus (Lemma 2.8); HL19 Rem. 3.7 only states the invariant case, with a sign slip.
(3) The bordered triangular system of HL19 Lemma 6.1 with its estimates (Lemma 2.12), and its exact origin as the reduction of the linearized isoenergetic equation in the frame (Lemma 2.14). *This is the only genuinely new analytic lemma*; HL19 print only "analogous to Lemma 5.1" for its proof, so the full proof is supplied.
(4) The approximate energy identities (Lemma 2.9): $D(H \circ K) = DH(K)DK$, $\partial_\omega(H \circ K) = -DH(K)E$, the compatibility $\langle \eta^N \rangle = 0$, and the size of the energy defect.
(5) Energy conservation of an invariant torus (Lemma 2.10): $E = 0$ and $\langle H \circ K \rangle = E_0$ give $H \circ K = E_0$ pointwise, the pointwise energy conclusion (O3) of I-G1.

Everything about the frame $L = DK$, N, P, Q, the symplectic defects and the cohomological equation is *cited* from track K's Part 2 ([K:`bp:p2:frame-identities`], [K:`bp:p2:exactness`], [K:`bp:p2:geometry`], [K:`bp:p2:cohomological`], [K:`bp:p2:triangular`], ...) and Part 1, not reproved: the canonical Case III objects of the frozen definitions are those of [K:`bp:p2:objects`] with h replaced by H.

2.1. Setting, sign dictionary, external facts.

Standing assumption 2.1 (Standing setting of Part 2)**.** Let $n \geq 2$ and $\tau \geq 0$ (I-G1 has $\tau \geq n-1$; only $\tau \geq 0$ is used here), $M > 0$. Let $D \subset \mathbb{C}^n \times \mathbb{C}^n$ be open with $(\theta + k, I) \in D$ for $(\theta, I) \in D$, $k \in \mathbb{Z}^n$ (I1); let $H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ be holomorphic on D, 1-periodic in θ, real at real phase points of D (I2, I3), with $\|DH\|, \|X_H\|, \|DX_H\|, \|D^2X_H\| \leq M$ on D (I4). Let $\gamma > 0$ and $\omega \in \mathbb{R}^n \cap \mathcal{D}_{\gamma,\tau}$. Let $r > 0$ and let $K : \mathbb{C}^n \to \mathbb{C}^n \times \mathbb{C}^n$ be holomorphic, primary and real on reals on T_r, with $K(T_r) \subset D$ (I5, I6). Where the frame is used, $\det G_K \neq 0$ on T_r.

This is exactly Assumption [K:`bp:p2:standing`] with $h := H$, $\mathfrak{B} := D$, together with the bound $\|DH\| \leq M$, which is used only in Lemma 2.9(e) and Lemma 2.5(b). All objects $L, G, B, N, P, X_N, T, E, A, \Omega_K, E_{\mathrm{sym}}, Q, \eta$ and the averages are those of [K:`bp:p2:objects`], with $h := H$; [K:`bp:p2:frozen-agreement`] identifies them with the frozen `gramMatrix`, `normalFrame`, `torsion`, `invarianceError`. The norms are those of Part 1: max norms, row-sum operator norms $|M| = \max_i \sum_j |M_{ij}|$ (so a $1 \times n$ matrix has the norm $\sum_j |M_{1j}|$, which is the norm of $DH(z)$ in (I4)). Write $\Gamma_\delta := c_R \gamma^{-1} \delta^{-\varkappa}$, $\varkappa = \tau + n$, as in [K:`bp:p2:constants`].

Remark 2.2 (Rays: $s\omega^*$ is Diophantine with the constant of ω^*). In I-G1, $\omega = s\omega^*$ with $s > 1$ and $\omega^* \in \mathcal{D}_{\gamma,\tau}$. For $k \neq 0$, $|k \cdot s\omega^*| = s|k \cdot \omega^*| \geq \gamma / \big(\sum_i |k_i|\big)^\tau$, so $\omega \in \mathcal{D}_{\gamma,\tau}$ with the *same* γ. Hence every result of K's Part 1 and Part 2 that requires $\omega \in \mathcal{D}_{\gamma,\tau}$ applies to $\omega = s\omega^*$ with Γ_δ as above.

Remark 2.3 (Lean totality for this part). Following Remark 1.13: every statement below that uses N, T, T_H assumes $\det G_K \neq 0$ on T_r first, so that $B = G^{-1}$ is the true inverse, holomorphic ([K:`bp:p1:holomorphic-inverse`]); every use of $\langle \mathsf{T}_H \rangle$ is preceded by continuity of T_H on $\mathbb{R}^n$ (Lemma 2.5(a)), hence integrability ([K:`bp:p1:ext-integral`]); every derivative of H, X_H is taken at a point of the open set D where H is holomorphic, so `fderiv` is the genuine derivative; the frozen T_H entry `fderiv` $\mathbb{C}$ H $(K\theta)$ $(N_{\cdot b})$ is then the matrix product $DH(K(\theta))N_{\cdot b}(\theta)$.

Remark 2.4 (Sign dictionary HL19 $\leftrightarrow$ this blueprint). HL19 use $\mathfrak{L}_\omega = -\sum_i \omega_i \partial_{\theta_i} = -\partial_\omega$ (4.1), p. 20, and the right operator $\mathfrak{R}_\omega$ of (4.2), $\widehat{(\mathfrak{R}_\omega v)}_k = -\hat{v}_k/(2\pi i k \cdot \omega)$. Track K's operator $\mathcal{R}$ ([K:`bp:p2:cohomological`]) solves $\partial_\omega u = g$, $\langle u \rangle = 0$, i.e. $\widehat{(\mathcal{R}g)}_k = \hat{g}_k/(2\pi i k \cdot \omega)$. Thus

$$\mathfrak{L}_\omega = -\partial_\omega, \qquad \mathfrak{R}_\omega = -\mathcal{R} \quad \text{(on the data of (4.1))}.$$

With the canonical $\Omega = \Omega_0$ and $\Omega X_h = (Dh)^\top$ (Rem. 2.1, p. 6; our [K:`bp:p2:hamiltonian`](d)), HL19's invariance error is $E = X_h(K) - DK\,\omega$ (2.2), p. 7, which p. 45 writes as $X_h(K) + \mathfrak{L}_\omega K$. The linearized system (6.1), p. 45, with $\bar{\omega} = \omega + \Delta\omega$, $\Delta\omega = -\omega\xi^\omega$ (6.3), is therefore

$$A\,\Delta K - \partial_\omega \Delta K - DK\,\Delta\omega = A\,\Delta K - \partial_\omega \Delta K + L\omega\,\xi^\omega = -E, \qquad \langle DH(K)\Delta K \rangle = -E^\omega,$$

$A = DX_H \circ K$, $L = DK$, $E^\omega = \langle H \circ K \rangle - E_0$, and $\bar{\omega} = (1 - \xi^\omega)\omega$ as on p. 50. The system (6.7) is

$$T\xi^N - \partial_\omega \xi^L + \omega\,\xi^\omega = \eta^L, \qquad -\partial_\omega \xi^N = \eta^N, \qquad \langle Y \xi^N \rangle = \eta^\omega,$$

and the solution (6.8)–(6.10) reads $\xi^N = \xi^N_0 - \mathcal{R}\eta^N$, $\xi^L = \xi^L_0 + \mathcal{R}\big(T\xi^N + \omega\xi^\omega - \eta^L\big)$, with (6.10) unchanged because $\mathfrak{R}_\omega = -\mathcal{R}$. Lemma 2.14 verifies that this is the equation obtained from (6.1) in the frame, signs included. Here and below $Y(\theta) := DH(K(\theta))N(\theta) \in \mathbb{C}^{1\times n}$ is HL19's H of (6.14), renamed to avoid a clash with the Hamiltonian.

External facts used in this part. Besides [K:`bp:p1:ext-integral`] (continuous functions on a compact box are integrable, integrals are linear and bounded by the supremum), the proofs use only:

- (X1) *Chain and product rules* for Fréchet derivatives of holomorphic maps, and linearity of the derivative (Mathlib `HasFDerivAt.comp`, `fderiv_add`, `fderiv_const`).
- (X2) *Complex derivative in a real direction.* If φ is holomorphic on an open $U \subset \mathbb{C}^n$, $x \in U \cap \mathbb{R}^n$ and $v \in \mathbb{R}^n$, then $\varphi|_{U\cap\mathbb{R}^n}$ has the real directional derivative $D\varphi(x)v$ at x; if moreover φ is real on $U \cap \mathbb{R}^n$, then $D\varphi(x)v \in \mathbb{R}$ (it is a limit of real difference quotients). See [K:`bp:p1:hamiltonian-real`], step [K:`step:p1:hr-derivative`].

2.2. **The extended torsion.**

Lemma 2.5 (Regularity, periodicity, realness and size of the extended torsion)**.** *Assume 2.1 with* $\det G_K \neq 0$ *on* T_r*, and put* $Y := DH(K)N$ *on* T_r*. Then:*

(a) Y *and the entries of* T_H *(built from* (H, K, ω)*) are holomorphic and* 1*-periodic on* T_r *and real at real* θ*. In particular* T_H *is continuous on* $\mathbb{R}^n$ *and* $\langle \mathsf{T}_H \rangle$ *is defined, real, and equals the block matrix*

$$\langle \mathsf{T}_H \rangle = \begin{pmatrix} \langle T \rangle & \omega \\ \langle Y \rangle & 0 \end{pmatrix}$$

with $\langle T \rangle = \langle T_{H,K,\omega} \rangle$ *the torsion average of* [K:`bp:p2:objects`].

(b) If $|DK| \le \sigma'_K$ *and* $|B| \le \sigma'_B$ *on* T_r*, then* $|Y| \le M\sigma'_K\sigma'_B$ *on* T_r*, and on any set where* $|T| \le C_T$*,* $|\mathsf{T}_H| \le \max\{C_T + |\omega|,\ M\sigma'_K\sigma'_B\}$ *in the row-sum norm of* $\mathbb{C}^{(n+1)\times(n+1)}$*.*

Frozen Lean declaration: `KAM.Iso.extendedTorsion, KAM.Iso.extendedTorsionAverage`
Source: HL19 (3.13), p. 19 ($c = H$*,* $d = n$*); HL19 (6.14), p. 47 for* Y*. Estimates: ours.*

Proof. **Step S2.1: Holomorphy.** N is holomorphic on T_r ([K:`bp:p2:frame-basic`](a)), $K(T_r) \subset D$ and DH is holomorphic on D ([K:`bp:p1:holomorphic-regularity`]), so $Y = DH \circ K \cdot N$ is a product of holomorphic functions ((X1)). The entries of T and the constant block ω are holomorphic

by [K:bp:p2:frame-basic](a). Hence every entry of T_H is holomorphic, in particular continuous on $\mathbb{R}^n \subset T_r$.

Step S2.2: Periodicity and realness. Differentiating $H(\theta+k, I) = H(\theta, I)$ gives $DH(z+(k,0)) = DH(z)$ on D, and $K(\theta+k) = K(\theta) + (k, 0)$; so $DH(K(\theta+k)) = DH(K(\theta))$. N and T are 1-periodic by [K:bp:p2:periodicity]. At real θ, $K(\theta)$ is a real phase point of D, where DH is real (the argument of [K:bp:p2:hamiltonian](c), step [K:step:p2:ham-real], applies to H, real at real points), and N, T are real ([K:bp:p2:periodicity]). Hence Y is real at real points.

Step S2.3: The average. By Step S2.1, each entry is integrable on $[0,1]^n$ ([K:bp:p1:ext-integral]), so the Bochner integral of the frozen $\langle \mathsf{T}_H \rangle$ is the entrywise integral, real because the integrand is real at real θ; the entries of the border are $\langle \omega \rangle = \omega$ (the cube has volume one) and $\langle 0 \rangle = 0$. (Lean totality, Remark 2.3.)

Step S2.4: Bounds. $|Y| \le |DH(K)||N| \le M\,|\Omega_0||L||B| \le M\sigma'_K\sigma'_B$, by [K:bp:p2:matrix](b), (d). The last column of T_H is the $n\times 1$ matrix ω, of norm $|\omega|$; by [K:bp:p2:matrix](c) the $(n+1)\times(n+1)$ norm is at most the maximum of the two block-row bounds $C_T + |\omega|$ and $|Y|$. □

Lemma 2.6 ($\mathsf{T}_H = \mathsf{B}_h$ at the trivial torus)**.** *Let $U \subset \mathbb{C}^n$ be open and $h : U \to \mathbb{C}$ holomorphic with $h(J) \in \mathbb{R}$ for every real $J \in U$. Let $D = \mathbb{C}^n \times U$, $H(\theta, J) := h(J)$ on D, $I \in U \cap \mathbb{R}^n$ and $K(\theta) = (\theta, I)$ ($\theta \in \mathbb{C}^n$). Put $\omega := \nabla h(I) \in \mathbb{R}^n$ (`frequencyMap`). Then K is holomorphic, primary and real on reals, $K(\mathbb{C}^n) \subset D$, $G_K = I_n$, $E_{H,K,\omega} \equiv 0$, and for every $\theta \in \mathbb{C}^n$*

$$\mathsf{T}_H(\theta) = \mathsf{B}_h(I) \quad \textit{(the real matrix, viewed in } \mathbb{C}^{(n+1)\times(n+1)}\textit{)}, \qquad \langle \mathsf{T}_H \rangle = \mathsf{B}_h(I),$$

the frozen `borderedHessian`. In particular $\det\langle \mathsf{T}_H\rangle \neq 0$ iff $\det \mathsf{B}_h(I) \neq 0$ and $|\langle \mathsf{T}_H\rangle^{-1}| = |\mathsf{B}_h(I)^{-1}|$.
Frozen Lean declaration: `KAM.Iso.extendedTorsion`, `KAM.Iso.extendedTorsionAverage`, `KAM.Iso.borderedHessian`
Source: ours; HL19 Rem. 3.7, p. 19, and Chi05 p. 17 (determinant $3/L^4$ in the Kepler case). Reading remark: Remark 1.17(b).

Proof. **Step S2.5: The torus.** $K(\theta+k) = (\theta+k, I) = K(\theta)+(k,0)$, K is affine, hence holomorphic, and real on real θ because I is real. $DK(\theta)e_i = (e_i, 0)$, so $G_K = (e_i \cdot e_j) = I_n$ (using the bilinear dot of Definition 1.5), $\det G_K = 1$ and $B = I_n$.

Step S2.6: The normal frame. $N_{\cdot j} = \sum_k \delta_{kj}\Omega_0(e_k, 0) = \Omega_0(e_j, 0) = (0, e_j)$, a constant function of θ, so $D_\theta N_{\cdot j} = 0$.

Step S2.7: The vector field. Since $H(\theta, J) = h(J)$, $X_H(\theta, J) = \big((\partial_a h(J))_a,\, 0\big)$, so $E_{H,K,\omega}(\theta) = X_H(K\theta) - DK\,\omega = (\nabla^{\mathbb{C}} h(I) - \omega, 0)$, where $\nabla^{\mathbb{C}} h(I) = (\partial_a h(I))_a$. By (X2) applied to $\varphi = h$ (real on $U \cap \mathbb{R}^n$), $\partial_a h(J) \in \mathbb{R}$ for real J, and the real function $J \mapsto \operatorname{Re} h(J) = h(J)$ on $U \cap \mathbb{R}^n$ has partial derivative $\partial_a h(J)$ in the direction e_a; by the definition of `frequencyMap`, $\omega_a = \partial_a h(I)$. Hence $E \equiv 0$, and more generally $\nabla h(J)_a = \partial_a h(J)$ for real $J \in U$.

Step S2.8: The torsion. $DX_H(\theta, I)(0, e_j) = \big((\partial_j\partial_a h(I))_a, 0\big)$, and $\Omega_0(w, 0) = (0, w)$. Hence $T(\theta)_{ij} = N_{\cdot i} \cdot \Omega_0\big(DX_H N_{\cdot j} - 0\big) = (0, e_i) \cdot (0, (\partial_j\partial_a h(I))_a) = \partial_j\partial_i h(I)$.

Step S2.9: The border. The last column is $\omega = \nabla h(I)$, entries $\omega_a = \partial_a h(I)$. The last row is $DH(K\theta)N_{\cdot b} = DH(\theta, I)(0, e_b) = \partial_b h(I) = \omega_b$. The corner is 0.

Step S2.10: Comparison with the bordered Hessian. The (a, b) entry of $\mathsf{B}_h(I)$ is $D(\nabla h)(I)e_b$ read at a, the real derivative in direction e_b of $J \mapsto \nabla h(J)_a = \partial_a h(J)$ on $U \cap \mathbb{R}^n$ (Step S2.7); since $\partial_a h$ is holomorphic, this is $\partial_b\partial_a h(I)$ by (X2), a real number. So the upper left blocks agree (and are symmetric by [K:bp:p1:holomorphic-regularity]); the border of $\mathsf{B}_h(I)$ is $\nabla h(I)$ in the last column and row, as found above. Thus $\mathsf{T}_H(\theta) = \mathsf{B}_h(I)$ for all θ; the constant has average equal to itself since $[0,1]^n$ has volume one. The statements on the determinant and the inverse

follow because a real matrix and its image in $\mathbb{C}^{(n+1)\times(n+1)}$ have the same determinant and inverse ([K:bp:p1:real-inverse]) and the same row-sum norm. □

Remark 2.7 (Use). For $H = h + f$ with f small, the torus $K(\theta) = (\theta, I)$ with $\omega = \nabla h(I)$ (or $s\omega^*$ with a nearby I, see Part 3) has $\langle \mathsf{T}_H \rangle = \mathsf{B}_h(I) + O(\|f\|)$ by Lemma 2.8 below and the Cauchy estimates; this is how (I8) of I-G1 is verified from $\det \mathsf{B}_h \neq 0$, with σ_T from [K:bp:p1:neumann].

Lemma 2.8 (The energy row of the extended torsion). *Assume 2.1 with* $\det G_K \neq 0$ *on* T_r *and let* $E = E_{H,K,\omega}$, $\eta^L = -N^\top \Omega_0 E$ *as in* [K:bp:p2:objects]. *Then on* T_r*:*

(a) $Y = DH(K)N = \omega^\top - (\eta^L)^\top$. *In particular* $|Y - \omega^\top| \le |\eta^L|$ *and, entrywise,* $Y_b = \omega_b + N_{\cdot b}^\top \Omega_0 E$.

(b) If $E = 0$ *on* T_r *(invariant torus), then* $Y = \omega^\top$ *on* T_r, *so*

$$\mathsf{T}_H(\theta) = \begin{pmatrix} T(\theta) & \omega \\ \omega^\top & 0 \end{pmatrix} \quad (\theta \in T_r), \qquad \langle \mathsf{T}_H \rangle = \begin{pmatrix} \langle T \rangle & \omega \\ \omega^\top & 0 \end{pmatrix}.$$

(c) In general $\langle \mathsf{T}_H \rangle = \mathsf{T}^{\mathrm{inv}} - \begin{pmatrix} 0 & 0 \\ \langle \eta^L \rangle^\top & 0 \end{pmatrix}$, *where* $\mathsf{T}^{\mathrm{inv}} := \left(\begin{smallmatrix} \langle T \rangle & \omega \\ \omega^\top & 0 \end{smallmatrix}\right)$ *is built from the same* T*; and if* $|E| \le e$ *on* T_r, $|L| \le \sigma'_K$, $|B| \le \sigma'_B$, *then* $|\langle \mathsf{T}_H \rangle - \mathsf{T}^{\mathrm{inv}}| \le 2n^2 \sigma'_K \sigma'_B \, e$.

Frozen Lean declaration: `KAM.Iso.extendedTorsion`

Source: HL19 Rem. 3.7, p. 19 (part (b), (3.17)), proved with the sign corrected (Remark 2.15, E3); (a), (c) are ours: HL19 (3.13) defines the last row as $Dc(K)N$ *without simplifying it off invariance.*

Proof. **Step S2.11: The gradient identity.** By [K:bp:p2:hamiltonian](d) (equivalently [K:bp:p1:hamiltonian-real]), $\Omega_0 X_H = (DH)^\top$ on D, so $DH(z) = (\Omega_0 X_H(z))^\top = X_H(z)^\top \Omega_0^\top = -X_H(z)^\top \Omega_0$.

Step S2.12: Insert the error. At $z = K(\theta)$, $X_H(K) = L\omega + E$ by the definition of E. Hence

$$Y = DH(K)N = -(L\omega + E)^\top \Omega_0 N = -\omega^\top L^\top \Omega_0 N - E^\top \Omega_0 N = \omega^\top - E^\top \Omega_0 N,$$

using $L^\top \Omega_0 N = -I_n$ ([K:bp:p2:frame-identities](a)). The b-th entry of the row $E^\top \Omega_0 N$ is the scalar $E^\top \Omega_0 N_{\cdot b} = N_{\cdot b}^\top \Omega_0^\top E = -N_{\cdot b}^\top \Omega_0 E = (\eta^L)_b$. This is (a) and the entrywise form. The row-sum norm of the $1 \times n$ matrix $(\eta^L)^\top$ is $\sum_b |\eta^L_b| \le n|\eta^L|$, so $|Y - \omega^\top| \le n|\eta^L|$.

Step S2.13: Invariant torus and average. (b) is (a) with $\eta^L = 0$ (since $E = 0$), together with the definition of T_H (Definition 1.6) and Lemma 2.5(a). For (c): averaging (a) and using linearity, $\langle Y \rangle = \omega^\top - \langle \eta^L \rangle^\top$, so the two matrices differ only in the last row, which has norm $\le n|\langle \eta^L \rangle| \le n \sup_{\mathbb{R}^n} |\eta^L|$ ([K:bp:p2:averages](a)), and $|\eta^L| \le |N^\top||\Omega_0||E| \le 2n\,|N||E| \le 2n\,\sigma'_K \sigma'_B \, e$ ([K:bp:p2:frame-basic](c), with $|N| \le |L||B|$). Hence the bound $2n^2 \sigma'_K \sigma'_B e$. □

2.3. **Energy identities.**

Lemma 2.9 (Approximate energy identities). *Assume 2.1 and let* $E = E_{H,K,\omega}$, $E^{\mathrm{en}} := H \circ K - \langle H \circ K \rangle$ *on* T_r. *Then:*

(a) $H \circ K$ *is holomorphic, 1-periodic on* T_r, *real on reals, and* $D(H \circ K) = DH(K)\, DK$; *the average* $\langle H \circ K \rangle = \bar{E}_H(K)$ *is the frozen* `energyAverage`, *real.*

(b) $\partial_\omega (H \circ K) = D(H \circ K)\omega = -DH(K)\, E$ *on* T_r.

(c) $L^\top \Omega_0 E = (D(H \circ K))^\top - \Omega_K \omega$ *on* T_r. *If* $\det G_K \neq 0$, *then* $\eta^N = L^\top \Omega_0 E$.

(d) $\langle D(H \circ K) \rangle = 0$, $\langle \Omega_K \rangle = 0$ *and* $\langle \eta^N \rangle = 0$ *(the compatibility condition of Lemma 2.12).*

(e) If $|E| \le \varepsilon$ *on* T_r *and* $0 < \delta < r$, *then* $|E^{\mathrm{en}}| \le M \Gamma_\delta \varepsilon$ *on* $T_{r-\delta}$, *and for* $0 < \delta' < r - \delta$, $|D(H \circ K)| \le nM\Gamma_\delta \varepsilon / \delta'$ *on* $T_{r-\delta-\delta'}$.

(f) (Energy defect.) Let $\det G_K \neq 0$ *on* T_r*, let* $\xi = (\xi^L, \xi^N)$ *be holomorphic on a strip* $T_{r'}$*,* $r' \leq r$*, and* $\Delta K = P\xi$*. Then on* $T_{r'}$

$$DH(K)\,\Delta K = Y\xi^N + D(H\circ K)\,\xi^L = Y\xi^N + (\eta^N)^\top \xi^L - \omega^\top \Omega_K\,\xi^L.$$

Consequently, if $\langle Y\xi^N\rangle = \eta^\omega$*, then* $\langle DH(K)\Delta K\rangle = \eta^\omega + E^\omega_{\mathrm{lin}}$ *with* $E^\omega_{\mathrm{lin}} := \langle D(H\circ K)\,\xi^L\rangle$*, and* $|E^\omega_{\mathrm{lin}}| \leq \sup_{\mathbb{R}^n}|D(H\circ K)|\,\sup_{\mathbb{R}^n}|\xi^L| \leq n\big(\sup|\eta^N| + |\omega|\sup|\Omega_K|\big)\sup|\xi^L|$.

Frozen Lean declaration: `KAM.Iso.energyAverage`

Source: HL19 Lemma 4.2, (4.3)–(4.5), p. 21, and (4.9) ($\mathfrak{L}_\omega(c\circ K) = Dc(K)E$*, equivalent to (b) with* $\mathfrak{L}_\omega = -\partial_\omega$*); p. 46 (*$Dc(K)L = D(c\circ K)$*,* E^ω_{lin} *of p. 49); (6.11)–(6.14), p. 47; compatibility* $\langle\eta^N\rangle = 0$ *from Lemma 4.6 as recalled on p. 47.*

Proof. **Step S2.14: Regularity and chain rule.** K is holomorphic on T_r with values in D, and H is holomorphic on D, so $H\circ K$ is holomorphic with $D(H\circ K) = DH(K)DK$ ((X1)). It is 1-periodic since $H(K(\theta+k)) = H(K(\theta)+(k,0)) = H(K(\theta))$, and real at real θ because $K(\theta)$ is a real phase point of D (I3). The average is $\int_{[0,1]^n} H(K(\theta))d\theta = \bar{E}_H(K)$ by definition, integrable by [K:`bp:p2:averages`](a). This is (a).

Step S2.15: The derivative along ω**.** $D(H\circ K)\omega = DH(K)L\omega = DH(K)\big(X_H(K) - E\big)$. By [K:`bp:p2:hamiltonian`](d), $DH = (\Omega_0 X_H)^\top$, so $DH(z)X_H(z) = X_H^\top\Omega_0^\top X_H = -X_H^\top\Omega_0 X_H = 0$ (Ω_0 is antisymmetric). Hence $D(H\circ K)\omega = -DH(K)E$, which is (b).

Step S2.16: The gradient identity. This repeats the gradient step of [K:`bp:p2:exactness`]: $L^\top\Omega_0 X_H(K) = (DH(K)L)^\top = (D(H\circ K))^\top$, hence $L^\top\Omega_0 E = (D(H\circ K))^\top - L^\top\Omega_0 L\omega = (D(H\circ K))^\top - \Omega_K\omega$. The identification $\eta^N = L^\top\Omega_0 E$ is the definition of η in [K:`bp:p2:objects`]. This is (c).

Step S2.17: Averages. $D(H\circ K)$ has the entries $\partial_{\theta_\ell}(H\circ K)$ of a holomorphic periodic function, so the average vanishes ([K:`bp:p2:averages`](b)); $\langle\Omega_K\rangle = 0$ and $\langle\eta^N\rangle = 0$ are [K:`bp:p2:exactness`]. (Also directly: $\langle\eta^N\rangle = \langle D(H\circ K)\rangle^\top + \langle\Omega_K\rangle\omega = 0$ by (c).)

Step S2.18: The size of the energy fluctuation. E^{en} is holomorphic and 1-periodic on T_r, has zero average, and by (b) $|\partial_\omega E^{\mathrm{en}}| = |DH(K)E| \leq |DH(K)||E| \leq M\varepsilon$ on T_r (here (I4), $K(T_r)\subset D$, is used). [K:`bp:p2:cohomological`](b), on T_r with loss δ, gives $|E^{\mathrm{en}}| \leq \Gamma_\delta M\varepsilon$ on $T_{r-\delta}$. The Cauchy estimate [K:`bp:p1:cauchy-strip`] for the bounded holomorphic function E^{en} on $T_{r-\delta}$ with loss δ' gives $|DE^{\mathrm{en}}| = |D(H\circ K)| \leq n\Gamma_\delta M\varepsilon/\delta'$ on $T_{r-\delta-\delta'}$. This is (e).

Step S2.19: The energy defect. $P = (L\ N)$, so $\Delta K = L\xi^L + N\xi^N$ and $DH(K)\Delta K = DH(K)L\xi^L + DH(K)N\xi^N = D(H\circ K)\xi^L + Y\xi^N$ by (a). By (c), $D(H\circ K) = (\eta^N + \Omega_K\omega)^\top = (\eta^N)^\top + \omega^\top\Omega_K^\top = (\eta^N)^\top - \omega^\top\Omega_K$ since $\Omega_K^\top = -\Omega_K$ ($\Omega_K = L^\top\Omega_0 L$, $\Omega_0^\top = -\Omega_0$). The average of the sum splits linearly. For the bound: $|\langle D(H\circ K)\xi^L\rangle| \leq \sup_{\mathbb{R}^n}|D(H\circ K)\xi^L|$ ([K:`bp:p2:averages`](a)), and $|D(H\circ K)| \leq |(\eta^N)^\top| + |\omega^\top\Omega_K| \leq n|\eta^N| + n|\omega||\Omega_K|$, with [K:`bp:p2:matrix`](a) for transposes (a $1\times n$ row $v^\top$ has $|v^\top| = \sum|v_i| \leq n|v|$). □

Lemma 2.10 (Energy conservation of an invariant torus)**.** *Assume 2.1 on a strip* T_r *(with* $\omega\in\mathcal{D}_{\gamma,\tau}$*) and* $E_{H,K,\omega}(\theta) = 0$ *for all* $\theta\in T_r$*. Then:*

(a) $H\circ K \equiv \bar{E}_H(K)$ *is constant on* T_r*.*
(b) If $E_0\in\mathbb{R}$ *(or* $\mathbb{C}$*) and* $\bar{E}_H(K) = E_0$*, then* $H(K(\theta)) = E_0$ *for every* $\theta\in T_r$*.*
(c) Quantitatively, if $E_{H,K,\omega}(\theta) = 0$ *on* T_r *then* $\sup_{T_r}|H\circ K - E_0| = |\bar{E}_H(K) - E_0|$*.*

Frozen Lean declaration: `KAM.Iso.energyAverage`

Source: HL19 Rem. 2.2, p. 7 ("quasi-periodicity implies $h(K(\theta)) = \langle h\circ K\rangle$*"); here proved. This is the pointwise energy conclusion (O3) of I-G1 (decision D-I-ENERGY).*

Proof. **Step S2.20: The derivative vanishes along ω.** Let $u := H \circ K - \langle H \circ K \rangle$, holomorphic and 1-periodic on T_r with $\langle u \rangle = 0$ (Lemma 2.9(a)). By Lemma 2.9(b), $Du\,\omega = -DH(K)E = 0$ on T_r.

Step S2.21: Uniqueness for the cohomological equation. For every $0 < \delta < r$, $u|_{T_{r-\delta}}$ is holomorphic, 1-periodic, of zero average, and solves $Du\,\omega = g$ with $g = 0$ (holomorphic, bounded, periodic, $\langle g \rangle = 0$ on T_r). By the uniqueness clause of [K:`bp:p1:cohomological`] (no boundedness of the solution assumed), $u = 0$ on $T_{r-\delta}$. As $\delta \in (0, r)$ is arbitrary, $u = 0$ on $T_r = \bigcup_\delta T_{r-\delta}$, which is (a).

Step S2.22: Pointwise energy. (b) and (c) are (a): $H \circ K \equiv \bar{E}_H(K)$, so $|H(K(\theta)) - E_0| = |\bar{E}_H(K) - E_0|$ for every θ. □

Remark 2.11 (Where (O3) of I-G1 comes from). In the proof of I-G1 (Part 2), $E_{H,K_\infty,s_\infty\omega^*} = 0$ on T_{ρ_∞} and $\bar{E}_H(K_\infty) = E_0$: the second follows from $|\bar{E}_H(K_j) - E_0| \to 0$ along the iteration and the uniform convergence $K_j \to K_\infty$ (continuity of H and of the integral on the compact cube), which is part of Part 2; then Lemma 2.10(b), with $\omega = s_\infty\omega^* \in \mathcal{D}_{\gamma,\tau}$ (Remark 2.2), gives $H(K_\infty) = E_0$ on T_{ρ_∞}. Neither the Gram determinant of K_∞ nor the frame is used (Remark 1.13(T2)).

2.4. **The bordered triangular system.** Notation: for $T : T_r \to \mathbb{C}^{n\times n}$ and $Y : T_r \to \mathbb{C}^{1\times n}$ holomorphic and 1-periodic put $\mathcal{T} := \begin{pmatrix} \langle T \rangle & \omega \\ \langle Y \rangle & 0 \end{pmatrix} \in \mathbb{C}^{(n+1)\times(n+1)}$; with $T = T_{H,K,\omega}$, $Y = DH(K)N$, this is $\langle \mathsf{T}_H \rangle$ (Lemma 2.5(a)). Let $\mathcal{R}$ be the solution operator of [K:`bp:p2:cohomological`](a): for g holomorphic, periodic with $\langle g \rangle = 0$, $\mathcal{R}g$ is the unique zero-average periodic solution of $\partial_\omega u = g$.

Lemma 2.12 (Bordered triangular equations (HL19 Lemma 6.1))**.** *Let $\gamma > 0$, $\tau \geq 0$, $\omega \in \mathbb{R}^n \cap \mathcal{D}_{\gamma,\tau}$, $r > 0$, $\delta > 0$ with $2\delta < r$, and $\Gamma_\delta = c_R\gamma^{-1}\delta^{-\varkappa}$. Let $\eta^L, \eta^N : T_r \to \mathbb{C}^n$ be holomorphic, 1-periodic, $|\eta^L| \leq H_L$ and $|\eta^N| \leq H_N$ on T_r, with $\langle \eta^N \rangle = 0$, and let $\eta^\omega \in \mathbb{C}$. Let $T : T_r \to \mathbb{C}^{n\times n}$ and $Y : T_r \to \mathbb{C}^{1\times n}$ be holomorphic and 1-periodic with $|T| \leq H_T$, $|Y| \leq H_Y$ on $T_{r-\delta}$. Assume $\det \mathcal{T} \neq 0$ and $|\mathcal{T}^{-1}| \leq \sigma$ (row-sum norm of $\mathbb{C}^{(n+1)\times(n+1)}$). Fix $\xi_0^L \in \mathbb{C}^n$ and define, in this order,*

$$
\begin{aligned}
u^N &:= \mathcal{R}\eta^N \quad \text{on } T_{r-\delta}, \\
\begin{pmatrix} \xi_0 \\ \xi^\omega \end{pmatrix} &:= \mathcal{T}^{-1} \begin{pmatrix} \langle \eta^L \rangle + \langle T u^N \rangle \\ \eta^\omega + \langle Y u^N \rangle \end{pmatrix} \in \mathbb{C}^n \times \mathbb{C}, \\
\xi^N &:= \xi_0 - u^N, \qquad g^L := T\xi^N + \omega\,\xi^\omega - \eta^L \quad \text{on } T_{r-\delta}, \\
\xi^L &:= \xi_0^L + \mathcal{R}g^L \quad \text{on } T_{r-2\delta}.
\end{aligned}
$$

Then:

(a) *(Well defined.) u^N, ξ^N, g^L are holomorphic and 1-periodic on $T_{r-\delta}$, $\langle g^L \rangle = 0$, so ξ^L is defined and holomorphic, 1-periodic on $T_{r-2\delta}$; the averages $\langle T u^N \rangle$, $\langle Y u^N \rangle$ exist.*

(b) *(The system.) On $T_{r-2\delta}$,*

$$T\xi^N - \partial_\omega \xi^L + \omega\,\xi^\omega = \eta^L, \qquad -\partial_\omega \xi^N = \eta^N, \qquad \langle Y\xi^N \rangle = \eta^\omega,$$

i.e. $\Lambda\xi - \partial_\omega \xi + \binom{\omega}{0}\xi^\omega = \eta$ with $\Lambda = \left(\begin{smallmatrix} 0 & T \\ 0 & 0 \end{smallmatrix}\right)$, $\xi = (\xi^L, \xi^N)$, $\eta = (\eta^L, \eta^N)$.

(c) *(Estimates.) With $S_* := \sigma \max\{H_L + H_T\Gamma_\delta H_N,\ |\eta^\omega| + H_Y\Gamma_\delta H_N\}$:*

$$|\xi_0|,\ |\xi^\omega| \leq |(\xi_0, \xi^\omega)| \leq S_*, \qquad \sup_{T_{r-\delta}} |\xi^N| \leq S_* + \Gamma_\delta H_N,$$

$$\sup_{T_{r-\delta}} |g^L| \leq H_T(S_* + \Gamma_\delta H_N) + |\omega| S_* + H_L, \qquad \sup_{T_{r-2\delta}} |\xi^L - \xi_0^L| \leq \Gamma_\delta \sup_{T_{r-\delta}} |g^L|.$$

(d) *(Uniqueness.) If $(\tilde{\xi}^L, \tilde{\xi}^N, \tilde{\xi}^\omega)$ are holomorphic, periodic on $T_{r-2\delta}$ ($\tilde{\xi}^\omega \in \mathbb{C}$) and satisfy the three equations of (b) on $T_{r-2\delta}$, then $\tilde{\xi}^\omega = \xi^\omega$, $\tilde{\xi}^N = \xi^N$ and $\tilde{\xi}^L - \xi^L$ is constant. Thus the solution is unique up to $\langle \xi^L \rangle$ (the free phase direction, HL19 Rem. 3.5).*

(e) (Realness.) If ω is real and η^L, η^N, T, Y are real on reals, $\eta^\omega \in \mathbb{R}$, $\mathcal{T}$ real (automatic), and $\xi_0^L \in \mathbb{R}^n$, then ξ_0, ξ^ω, ξ^N, ξ^L, g^L are real on reals.

Source: HL19 Lemma 6.1, pp. 46–47, (6.7)–(6.10), with signs converted by Remark 2.4; the estimates are those on p. 46, with the additional term $|\omega|S_*$ in g^L (see Remark 2.13). Structure as in [K:bp:p2:triangular] *(HL19 Lemma 5.1, pp. 31–32).*

Source gap: HL19 prove Lemma 6.1 in four lines ("It is analogous to Lemma 5.1…"), and Lemma 5.1 is itself only referred to ("directly adapted from Lemma 4.14 in [33]"). The proof below is supplied here in full; the new points relative to [K:bp:p2:triangular] *are Steps S2.25, S2.26 and S2.27, and uniqueness (d).*

Proof. **Step S2.23: First cohomological equation.** η^N is holomorphic, periodic on T_r, bounded by H_N, with $\langle \eta^N \rangle = 0$. [K:bp:p2:cohomological](a) gives $u^N = \mathcal{R}\eta^N$ holomorphic and 1-periodic on $T_{r-\delta}$, $\langle u^N \rangle = 0$, $\partial_\omega u^N = \eta^N$, $|u^N| \le \Gamma_\delta H_N$ on $T_{r-\delta}$, real on reals if η^N is.

Step S2.24: Averages of the products. Tu^N and Yu^N are holomorphic and periodic on $T_{r-\delta}$, hence continuous on $\mathbb{R}^n$, so their averages exist ([K:bp:p2:averages](a)) and $|\langle Tu^N \rangle| \le H_T \Gamma_\delta H_N$, $|\langle Yu^N \rangle| \le H_Y \Gamma_\delta H_N$, $|\langle \eta^L \rangle| \le H_L$. Let $w = (\langle \eta^L \rangle + \langle Tu^N \rangle,\ \eta^\omega + \langle Yu^N \rangle) \in \mathbb{C}^{n+1}$; in the max norm of $\mathbb{C}^{n+1}$, $|w| \le \max\{H_L + H_T\Gamma_\delta H_N, |\eta^\omega| + H_Y \Gamma_\delta H_N\}$.

Step S2.25: The bordered average system. *(new relative to* [K:bp:p2:triangular]*.)* Since $\det \mathcal{T} \neq 0$, $(\xi_0, \xi^\omega) := \mathcal{T}^{-1} w$ is the unique solution of

$$\langle T \rangle \xi_0 + \omega\, \xi^\omega = \langle \eta^L \rangle + \langle Tu^N \rangle, \qquad \langle Y \rangle \xi_0 = \eta^\omega + \langle Yu^N \rangle, \tag{1}$$

and $|(\xi_0, \xi^\omega)| \le |\mathcal{T}^{-1}||w| \le S_*$ ([K:bp:p2:matrix](b)). This is (6.10) of HL19 in our sign convention (Remark 2.4). Note that $\langle T \rangle$ alone need not be invertible: the matrix $\mathcal{T}$ determines the constant ξ_0 and the frequency correction ξ^ω together.

Step S2.26: The energy equation and ξ^N. By Step S2.23, $\xi^N = \xi_0 - u^N$ is holomorphic, periodic on $T_{r-\delta}$ with $\partial_\omega \xi^N = -\eta^N$ and $|\xi^N| \le |\xi_0| + \Gamma_\delta H_N \le S_* + \Gamma_\delta H_N$. By linearity of the average and the second equation of (1), $\langle Y \xi^N \rangle = \langle Y \rangle \xi_0 - \langle Yu^N \rangle = \eta^\omega$. This is the third equation of (b) and the second one.

Step S2.27: The second cohomological equation. *(new relative to* [K:bp:p2:triangular]*.)* $g^L = T\xi^N + \omega\xi^\omega - \eta^L$ is holomorphic and periodic on $T_{r-\delta}$ with $|g^L| \le H_T(S_* + \Gamma_\delta H_N) + |\omega| S_* + H_L$. Its average is, by linearity, $\langle T \rangle \xi_0 - \langle Tu^N \rangle + \omega\xi^\omega - \langle \eta^L \rangle = 0$ by the first equation of (1). This is the point of the border: the frequency correction ξ^ω absorbs the part of the mean of $T\xi^N - \eta^L$ that the constant ξ_0 cannot adjust once it is spent on the energy equation. [K:bp:p2:cohomological](a) on $T_{r-\delta}$ with loss δ (valid since $\delta < r - \delta$) gives $\mathcal{R}g^L$ holomorphic, periodic on $T_{r-2\delta}$, with $\partial_\omega \mathcal{R} g^L = g^L$ and $|\mathcal{R}g^L| \le \Gamma_\delta \sup|g^L|$. Then $T\xi^N - \partial_\omega \xi^L + \omega\xi^\omega = T\xi^N - g^L + \omega\xi^\omega = \eta^L$. With Steps S2.23 and S2.26 this is (a), (b), (c).

Step S2.28: Uniqueness. Let $(\tilde\xi^L, \tilde\xi^N, \tilde\xi^\omega)$ be as in (d). Then $\partial_\omega(\tilde\xi^N + u^N) = -\eta^N + \eta^N = 0$ on $T_{r-2\delta}$, and $\tilde\xi^N + u^N - \tilde c$, $\tilde c := \langle \tilde\xi^N + u^N \rangle$, is a zero-average holomorphic periodic solution of $Du\,\omega = 0$; by the uniqueness in [K:bp:p1:cohomological] (as in Step S2.21) it is 0. So $\tilde\xi^N = \tilde c - u^N$. Averaging the first equation of (b) for the tilde-solution ($\langle \partial_\omega \tilde\xi^L \rangle = 0$, [K:bp:p2:averages](b)) and using the third gives exactly (1) for $(\tilde c, \tilde\xi^\omega)$, whence $(\tilde c, \tilde\xi^\omega) = (\xi_0, \xi^\omega)$ by invertibility of $\mathcal{T}$, and $\tilde\xi^N = \xi^N$. Then $\partial_\omega(\tilde\xi^L - \xi^L) = 0$, so $\tilde\xi^L - \xi^L$ is constant, by the same uniqueness lemma applied to $\tilde\xi^L - \xi^L - \langle \tilde\xi^L - \xi^L \rangle$.

Step S2.29: Realness. If the data are real on reals, $\langle T \rangle, \langle Y \rangle, \langle \eta^L \rangle, \langle Tu^N \rangle, \langle Yu^N \rangle$ are real ([K:bp:p2:averages](a)), $\mathcal{T}$ is a real matrix with real inverse ([K:bp:p1:real-inverse]), so ξ_0, ξ^ω are real; $\mathcal{R}$ preserves realness ([K:bp:p2:cohomological](a)), so u^N, ξ^N, g^L, ξ^L are real on reals. □

Remark 2.13 (Differences from HL19's printed statement). (i) HL19's $\mathfrak{R}_\omega$ discards the average of its argument, so their (6.9) $\xi^L = \xi^L_0 + \mathfrak{R}_\omega(\eta^L - T\xi^N)$ omits the constant vector $\omega\xi^\omega$, which has no effect on non-zero Fourier modes; in the operator $\mathcal{R}$ of [K:bp:p2:cohomological] (defined on zero-average data) the term is carried explicitly in g^L, which costs the extra summand $|\omega|S_*$ in the bound of $\sup|g^L|$. HL19's bound for ξ^L (p. 46, last line) is true but is not obtained from [K:bp:p2:cohomological] as stated; we do not need it, because $|\omega| \le \sigma_\omega|\omega^*|$ is available in the application. (ii) HL19 have real data and $\xi^L_0 \in \mathbb{R}^n$; our lemma holds for complex data and records realness separately. (iii) The estimates of HL19 are in max of the two $\mathbb{C}^{n+1}$ block components, as here. (iv) Uniqueness (d) is not stated in HL19.

Lemma 2.14 (The linearized isoenergetic equation in the frame)**.** *Assume 2.1 with* $\det G_K \neq 0$ *on* T_r *and* $\omega \in \mathbb{R}^n$ *(any). Let* $\xi = (\xi^L, \xi^N)$ *be holomorphic on* $T_{r'}$, $r' \le r$, *and* $\xi^\omega \in \mathbb{C}$*; put* $\Delta K = P\xi$, $\Delta\omega = -\omega\xi^\omega$. *Then on* $T_{r'}$*:*

(a) $QL = \binom{I_n}{-\Omega_K}$, *hence* $QL\omega\,\xi^\omega = \binom{\omega}{-\Omega_K\omega}\xi^\omega$.

(b) $Q\big(A\,\Delta K - \partial_\omega \Delta K - DK\,\Delta\omega\big) + QE = \Lambda\xi - \partial_\omega\xi + \binom{\omega}{0}\xi^\omega - \eta + E_{\mathrm{lin}}$, *where*

$$E_{\mathrm{lin}} := E_{\mathrm{red}}\xi + \Omega_0 E_{\mathrm{sym}}\,\partial_\omega\xi - \begin{pmatrix} 0 \\ \Omega_K\omega \end{pmatrix}\xi^\omega.$$

Consequently $A\Delta K - \partial_\omega\Delta K + L\omega\xi^\omega = -E$ *holds exactly when the bordered system of Lemma 2.12(b) holds up to* E_{lin}, *i.e. when* $\Lambda\xi - \partial_\omega\xi + \binom{\omega}{0}\xi^\omega = \eta - E_{\mathrm{lin}}$*; the data are* η^L, η^N *of* [K:bp:p2:objects], T *the torsion, and* $Y = DH(K)N$. *In particular, with* ξ *the solution of Lemma 2.12 (whose first two equations give* $\Lambda\xi - \partial_\omega\xi + \binom{\omega}{0}\xi^\omega = \eta$*), the residual of the linearized equation is* $Q^{-1}E_{\mathrm{lin}}$, *and* $|E_{\mathrm{lin}}| \le |E_{\mathrm{red}}||\xi| + |E_{\mathrm{sym}}||\partial_\omega\xi| + |\Omega_K||\omega||\xi^\omega|$, *all quadratic in* (E, ξ) *by* [K:bp:p2:geometry].
Source: HL19 (6.4)–(6.5), p. 45, with $\Omega = \Omega_0$, $A = 0$ *(canonical Case III); (6.34), p. 49 for* E_{lin}. *Frame identities:* [K:bp:p2:frame-identities].

Proof. **Step S2.30: The product** QL**.** $Q = -\Omega_0 P^\top \Omega_0$ and $P^\top\Omega_0 L = \binom{L^\top\Omega_0 L}{N^\top\Omega_0 L} = \binom{\Omega_K}{I_n}$ by [K:bp:p2:frame-identities](a). By [K:bp:p2:matrix](d), $\Omega_0\binom{U_1}{U_2} = \binom{-U_2}{U_1}$, so $QL = -\binom{-I_n}{\Omega_K} = \binom{I_n}{-\Omega_K}$. This is (a).

Step S2.31: Expanding the linearized operator. $\partial_\omega(P\xi) = (\partial_\omega P)\xi + P\partial_\omega\xi$, so $A\Delta K - \partial_\omega\Delta K = (AP - \partial_\omega P)\xi - P\partial_\omega\xi$, and $-DK\,\Delta\omega = L\omega\,\xi^\omega$. By [K:bp:p2:frame-identities](e), $Q(AP - \partial_\omega P) = \Lambda + E_{\mathrm{red}}$; by (f), $QP = I_{2n} - \Omega_0 E_{\mathrm{sym}}$; by (a), $QL\omega\xi^\omega = \binom{\omega}{0}\xi^\omega - \binom{0}{\Omega_K\omega}\xi^\omega$. Adding $QE = -\eta$: $Q(\dots) + QE = (\Lambda + E_{\mathrm{red}})\xi - \partial_\omega\xi + \Omega_0 E_{\mathrm{sym}}\partial_\omega\xi + \binom{\omega}{0}\xi^\omega - \binom{0}{\Omega_K\omega}\xi^\omega - \eta$, which is (b).

Step S2.32: Consequence. If the border system holds, (b) gives $Q\,\mathrm{Res} = E_{\mathrm{lin}}$ for the residual $\mathrm{Res} := A\Delta K - \partial_\omega\Delta K + L\omega\xi^\omega + E$. $Q = -\Omega_0 P^\top\Omega_0$ is invertible iff P is, and $\det(P)^2 \det\Omega_0 = \det(P^\top\Omega_0 P) = \det(\Omega_0 + E_{\mathrm{sym}}) \neq 0$ when $|E_{\mathrm{sym}}| < 1$ (Neumann series [K:bp:p1:neumann] applied to $I - \Omega_0 E_{\mathrm{sym}}$, $|\Omega_0 E_{\mathrm{sym}}| = |E_{\mathrm{sym}}|$, [K:bp:p1:matrix-norms]). The bound on E_{lin} is [K:bp:p2:matrix](b), (c). □

2.5. **Suspected errata and sign checks.**

Remark 2.15 (Candidate errata of HL19, checked against the pinned PDF (arxiv_1711.05515.pdf, v1)). All three were checked on the rendered pages 18, 19, 46 (not only on the text layer).

E1 (strict inequalities).: Thm 3.6, (3.16), p. 19: "$\|K_\infty - K\|_{\rho_\infty} < \dfrac{\mathfrak{C}_2\|E_c\|_\rho}{\gamma^2\rho^{2\tau}}$, $|\omega_\infty - \omega| < \dfrac{\mathfrak{C}_3\|E_c\|_\rho}{\gamma^2\rho^{2\tau}}$"; likewise (3.11), p. 18 for Thm 3.1. With $E_c \equiv 0$ the right side is 0, and (3.15) holds, but no K_∞ satisfies $\|K_\infty - K\| < 0$. The statements must read "$\le$" (the frozen I-G1 does).

E2 (sign in (3.12)).: p. 18: "(3.12) $X_h(K(\theta)) = \mathfrak{L}_\omega K(\theta)$, $\langle c \circ K\rangle = c_0$". With $\mathfrak{L}_\omega = -\partial_\omega$ (4.1), p. 20, this reads $X_h(K) = -DK\omega$, contradicting the invariance equation (2.1), p. 6, $X_h(K) = DK\,\omega$, and the error (2.2), p. 7, $E = X_h(K) - DK\omega$. The correct form is $X_h(K) + \mathfrak{L}_\omega K = 0$, which is how p. 45 (the total error before (6.1)) and p. 48 (Lemma 6.2, $\bar{E}_c$) write it. Our Remark 2.4 uses the consistent convention.

E3 (Rem. 3.7, first equality).: p. 19: "$Dh(K(\theta))N(\theta) = X_h(K(\theta))^\top \Omega(K(\theta))N(\theta) = -\omega^\top DK(\theta)^\top \Omega(K(\theta))N(\theta) = (\omega^\top\ 0_{n-d}^\top) = \hat{\omega}^\top$". By Rem. 2.1, p. 6, $\Omega X_h = (Dh)^\top$, so $Dh = (\Omega X_h)^\top = -X_h^\top \Omega$: the first equality misses a minus sign. The second equality is likewise wrong for $X_h = +DK\omega$ (it would need $X_h = -DK\omega$, the E2 convention). The two slips cancel and the final value $Dh\,N = \hat{\omega}^\top$ is correct, as Lemma 2.8(a) shows (using $L^\top \Omega_0 N = -I_n$, [K:`bp:p2:frame-identities`](a)); an independent check at the trivial torus is Lemma 2.6, Step S2.9. The correct chain is $Dh\,N = -X_h^\top \Omega N = -\omega^\top DK^\top \Omega N = \omega^\top$.

Other points checked and found consistent (no erratum): (6.7)–(6.10), p. 46, with $\mathfrak{L}_\omega = -\partial_\omega$, $\mathfrak{R}_\omega = -\mathcal{R}$ (Remark 2.4); (6.3) and $\bar{\omega} = (1 - \xi^\omega)\omega$ (p. 50, (6.32)); the equation $QL\omega = \binom{\omega}{0} - \binom{0}{\Omega_K \omega}$ against the display on p. 46 (Lemma 2.14(a)); the sign $\Delta\omega = -\omega\xi^\omega$. HL19's (6.9) omits the constant $\omega\xi^\omega$ (Remark 2.13(i)), which is harmless.

Remark 2.16 (Doubts about the frozen statements). None of the frozen statements used in this part looks false. Two reading points. (a) The frozen B_h uses the real derivative of $\operatorname{Re} h$ via `frequencyMap`; Lemma 2.6 identifies it with the complex Hessian of h only under "h real on the real points of a neighbourhood of I", which I-HEAD assumes on all of B_{r_0}; without that hypothesis T_h and B_h can differ by the imaginary part. (b) The frozen last row of T_H is $DH(K)N$, not $\omega^\top$ (Definition 1.6); Lemma 2.8 shows the difference is $(\eta^L)^\top = O(e)$, so the two readings give the same I-G1 up to the choice of σ_T.

3. One isoenergetic step, the iteration, and the proof of I-G1

This part proves Theorem 1.14 (I-G1). It adapts the quasi-Newton scheme of track K's Part 2 ([K:`bp:p2:standing`]–[K:`bp:p2:g1`]) to HL19's isoenergetic scheme (§6.2–6.3, pp. 47–53). The unknown of one step is now the pair $(\Delta K, \Delta\omega)$ with $\Delta\omega$ *parallel to the ray* $\mathbb{R}\omega^*$, determined together with ΔK by the bordered system of Part 2 (Lemma 2.12): the frequency correction ξ^ω absorbs the mean of the torsion equation that the constant ξ_0 cannot, once ξ_0 is spent on the energy equation $\langle DH(K)\Delta K\rangle = -E^\omega$.

What changes with respect to track K.

(1) The frequency of the s-th iterate is $\omega_s = s_s\omega^*$ with s_s real and moving; the Diophantine condition is inherited from ω^* (Remark 2.2). The distance to the two endpoints of the segment (hypothesis (I9), reading remark 1.15(b)) is carried as a margin that decreases from ν to at least $\nu/2$.

(2) The invariance error and the energy error $E^\omega = \langle H \circ K\rangle - E_0$ are controlled together, and the averaged $(n+1)\times(n+1)$ extended torsion replaces $\langle T\rangle$. Its inverse is perturbed by the Neumann series ([K:`bp:p1:neumann`]), which is HL19 Lemma A.1, p. 55, in the max-row-sum norm.

(3) *Open point of the plan: does* [K:`bp:p2:newton-error`] *apply?* No, and not merely for lack of hypotheses (see Remark 3.9): for the frozen frequency ω the first bracket $X_H(K+\Delta K) - D(K+\Delta K)\omega$ has a *first-order* defect $L\Delta\omega$, removed only by the change of frequency. The new error is therefore computed at $\bar{\omega} = \omega + \Delta\omega$ in Lemma 3.8, with the extra term $-D\Delta K\,\Delta\omega$ (HL19 p. 50, $L_{\Delta\omega}\Delta K$).

(4) Several lemmas of track K are stated for tori that are (r, β, ε)-admissible, a notion that includes invertibility of the *torsion* average $\langle T\rangle$ (condition [K:`bp:p2:A5`]). In the isoenergetic

setting $\langle T\rangle$ itself need not be invertible, so those lemmas are not cited; their conclusions that do not involve $\langle T\rangle$ are restated (Lemmas 3.5, 3.6) and proved without that hypothesis.

No other result of track K is changed. Constants depend on $(n,\tau,M,\sigma_K,\sigma_B,\sigma_T,\mu,\nu)$ only, *not* on ω^*, σ_ω, $\bar\gamma$ (except through C_1, see Theorem 3.17); the hypotheses (I9), (I10) are used qualitatively.

3.1. **Standing data, the ray, constants.**

Standing assumption 3.1 (Standing data of Part 3)**.** Fix $n\ge 2$, $\tau\ge n-1$, $M,\sigma_K,\sigma_B,\sigma_T>0$, $\sigma_\omega>1$, $\nu>0$, $\omega^*\in\mathbb{R}^n\setminus\{0\}$. Let $D\subset\mathbb{C}^n\times\mathbb{C}^n$ and $H:\mathbb{C}^n\times\mathbb{C}^n\to\mathbb{C}$ satisfy (I1)–(I4) of Theorem 1.14 (open translation-invariant D; H holomorphic, 1-periodic in θ, real at real points of D; $\|DH\|,\|X_H\|,\|DX_H\|,\|D^2X_H\|,\|D^2H\|\le M$ on D). Let $E_0\in\mathbb{R}$, $\gamma>0$ and $\omega^*\in\mathcal{D}_{\gamma,\tau}$.

Taking $\mathfrak{B}:=D$, $h:=H$ this contains Assumption [K:`bp:p2:standing`] (which does not use $\|DH\|\le M$ or $\|D^2H\|\le M$); hence [K:`bp:p2:hamiltonian`] applies to H on D, and Assumption 2.1 holds for every $\omega=s\omega^*$ with $s>1$. All objects $L,G,B,N,P,A,T,E,\Omega_K,E_{\mathrm{sym}},E_{\mathrm{red}},Q,\eta$ are those of [K:`bp:p2:objects`] with $h:=H$; $Y:=DH(K)N$; $\mathcal{T}:=\langle\mathsf{T}_H\rangle$. The energy error of a torus K is $E^\omega(K):=\bar E_H(K)-E_0=\langle H\circ K\rangle-E_0$.

Lemma 3.2 (Frequencies on the ray)**.** *Let $s,\bar s\in\mathbb{R}$ and $\omega=s\omega^*$, $\bar\omega=\bar s\omega^*$.*

(a) If $s>0$, then $\omega\in\mathcal{D}_{\gamma s,\tau}$; if $s\ge 1$, then $\omega\in\mathcal{D}_{\gamma,\tau}$. Moreover $\gamma\le|\omega_1^*|\le|\omega^*|$, and $|\bar\omega-\omega|=|\bar s-s||\omega^*|$.

(b) Put $\nu(s):=\min\{s-1,\sigma_\omega-s\}|\omega^*|$. Then $\nu(\bar s)\ge\nu(s)-|\bar\omega-\omega|$. In particular, if $\nu(s)\ge\nu'$ and $|\bar\omega-\omega|\le d<\nu'$, then $1<\bar s<\sigma_\omega$ and $\nu(\bar s)\ge\nu'-d>0$.

Source: HL19 (H_5'), p. 19 and (6.22), p. 48; ours.

Proof. **Step S3.1: Diophantine.** For $k\ne 0$, $|k\cdot s\omega^*|=|s||k\cdot\omega^*|\ge|s|\gamma(\sum|k_i|)^{-\tau}$ (Remark 2.2 for $s\ge1$). Taking $k=e_1$ gives $\gamma\le|\omega_1^*|$.

Step S3.2: Margins. $\bar s-1\ge s-1-|\bar s-s|$ and $\sigma_\omega-\bar s\ge\sigma_\omega-s-|\bar s-s|$; multiply by $|\omega^*|$. □

Definition 3.3 (Constants)**.** Let $\varkappa,a,c_R,\mathfrak{m},\Theta,C_X,C_T,C_\Omega,C_{\mathrm{sym}},C_{\mathrm{red}},C_{\xi_0},C_G,C_B,C_{\Delta N},C_{\Delta X},C_{\Delta T}$ be the constants of [K:`bp:p2:constants`]; they depend only on $(n,\tau,M,\sigma_K,\sigma_B,\sigma_T)$, and $\varkappa=\tau+n\ge 2n-1\ge3$, $a=2\varkappa+2\ge8$. Define, in this order,

$$\hat C_S:=\max\{C_{\xi_0},\ \mathfrak{m}\Theta+2nc_R\mathfrak{m}^5\},\qquad \hat C_{\xi N}:=\hat C_S+2nc_R\mathfrak{m},$$
$$\hat C_g:=C_T\hat C_{\xi N}+\Theta\hat C_S+2n\mathfrak{m}^2\Theta,\qquad \hat C_\xi:=\max\{c_R\hat C_g,\ \Theta\hat C_{\xi N}\},$$
$$\hat C_{\Delta K}:=2\mathfrak{m}^2\hat C_\xi,\qquad \hat C_\Delta:=\max\{\hat C_{\Delta K},\ \Theta^2\hat C_S\},$$
$$C_P:=\mathfrak{m}^2+n+\mathfrak{m}^3+C_X,\qquad \hat C_\partial:=C_P\hat C_\xi+2\mathfrak{m}^2\Theta\hat C_g,$$
$$\hat C_{\mathrm{lin}}:=C_{\mathrm{red}}\hat C_\xi+C_{\mathrm{sym}}\hat C_g+\Theta C_\Omega\hat C_S,\qquad \hat C_{E1}:=4\mathfrak{m}^2\Theta^2\hat C_{\mathrm{lin}}+\mathfrak{m}\hat C_{\Delta K}^2+\Theta\hat C_S\hat C_\partial,$$
$$\hat C_{E2}:=n\Theta^2(2n\mathfrak{m}+C_\Omega)c_R\hat C_g+\mathfrak{m}\hat C_{\Delta K}^2,\qquad \hat C_E:=\max\{\hat C_{E1},\hat C_{E2}\},$$
$$C_Y:=n\mathfrak{m}(2\mathfrak{m}^2+C_{\Delta N}),\qquad C_{\Delta\mathsf{T}}:=C_{\Delta T}+8n^2\mathfrak{m}^4+1+C_Y,$$
$$\hat{\mathfrak{c}}_{\mathrm{step}}:=2\Theta^3\max\{C_{\mathrm{sym}},\ \mathfrak{m}C_{\Delta\mathsf{T}}\hat C_\Delta\},\qquad \hat{\mathfrak{c}}_{\mathrm{sum}}:=4\Theta^2\hat C_\Delta\max\{n,\ C_B,\ \mathfrak{m}^2C_{\Delta\mathsf{T}}\}.$$

For $\mu,\nu>0$ and $0<\rho_\infty<\rho$ put

$$\hat{\mathfrak{c}}_*:=\max\Big\{2^{2a}\hat C_E,\ \hat{\mathfrak{c}}_{\mathrm{step}}\max\{1,2/\mu\},\ \frac{2\hat{\mathfrak{c}}_{\mathrm{sum}}}{\mu},\ \frac{4\Theta^2\hat C_\Delta}{\nu}\Big\},\qquad \delta_0:=\min\Big\{1,\frac{\rho-\rho_\infty}{8}\Big\}.$$

All are finite, ≥1 unless stated otherwise, depend only on $(n,\tau,M,\sigma_K,\sigma_B,\sigma_T)$, and $\hat{\mathfrak{c}}_*$ in addition on (μ,ν), δ_0 on (ρ,ρ_∞). Each is introduced in one result (register, Remark 3.18).

Source: HL19 Table 4, Table 5, p. 59, (6.45)–(6.51), pp. 52–53 (their constants depend on the current torus; here, as in [K:`bp:p2:constants-register`], through $\mathfrak{m}$ and the margin μ).

Definition 3.4 (Isoenergetic admissible parameterization)**.** Let $r, \beta > 0$, $0 \le \varepsilon \le 1$, $s > 1$, $\omega := s\omega^*$. A map $K : \mathbb{C}^n \to \mathbb{C}^n \times \mathbb{C}^n$ is $(r, \beta, \varepsilon, s)$-*admissible* if

(B1) K is holomorphic on T_r, primary on T_r, and real on reals;
(B2) $B_\beta(K(\theta)) \subset D$ for $\theta \in T_r$;
(B3) $|DK| \le \sigma_K$ on T_r;
(B4) $\det G_K \neq 0$ and $|B| \le \sigma_B$ on T_r;
(B5) $\det\langle \mathsf{T}_H \rangle \neq 0$ (T_H built from (H, K, ω); defined by Lemma 2.5) and $|\langle \mathsf{T}_H \rangle^{-1}| \le \sigma_T$;
(B6) $|E_{H,K,\omega}| \le \varepsilon$ on T_r;
(B7) $|\bar{E}_H(K) - E_0| \le \varepsilon$.

For such K the frequency bound of [K:bp:p2:frequency] (hypotheses [K:bp:p2:A3], [K:bp:p2:A4], [K:bp:p2:A6] are (B3), (B4), (B6); $\omega \in \mathcal{D}_{\gamma,\tau}$ by Lemma 3.2) gives

$$\gamma \le |\omega| \le \Theta, \tag{2}$$

and $\mathfrak{m}, \Theta, \gamma$ satisfy $\gamma\delta^j \le \Theta$ for $\delta \le 1$, $j \ge 0$: absorption steps below use this exactly as [K:step:p2:cor-absorb].

The hypotheses (I5)–(I12) of I-G1 say that K is (ρ, μ, e, s)-admissible with the sharper constants $\sigma_K - \mu, \sigma_B - \mu, \sigma_T - \mu$ in (B3)–(B5), once $e \le 1$ is known (Step S3.40).

3.2. Geometry and frame perturbation without the torsion hypothesis.

Lemma 3.5 (Geometry of an approximately invariant torus, without [K:bp:p2:A5])**.** *Let $\omega \in \mathcal{D}_{\gamma,\tau}$, $0 < \delta \le 1$, $2\delta < r$, $0 \le \varepsilon \le 1$, and let K satisfy (B1)–(B4), (B6) on T_r (no condition on any torsion average). Then the conclusions of* [K:bp:p2:geometry] *hold: $|DE| \le n\varepsilon/\delta$, $|\partial_\omega \Omega_K| \le 2n(n+1)\mathfrak{m}\varepsilon/\delta$, $|E_{\mathrm{red}}| \le C_{\mathrm{red}}\varepsilon/\delta$ on $T_{r-\delta}$, and $|\Omega_K| \le C_\Omega \varepsilon/(\gamma\delta^{\varkappa+1})$, $|E_{\mathrm{sym}}| \le C_{\mathrm{sym}}\varepsilon/(\gamma\delta^{\varkappa+1})$ on $T_{r-2\delta}$. Also, on T_r: $|\eta^L| \le 2n\mathfrak{m}^2\varepsilon$, $|\eta^N| \le 2n\mathfrak{m}\varepsilon$ and $|L| \le \mathfrak{m}$, $|B| \le \mathfrak{m}$, $|N| \le \mathfrak{m}^2$, $|P| \le 2\mathfrak{m}^2$; and if $|\omega| \le \Theta$, then $|X_N| \le C_X/\delta$, $|T| \le C_T/\delta$ on $T_{r-\delta}$.*
Source: HL19 Lemma 4.3, 4.6, 4.10, pp. 22–30, as in [K:bp:p2:geometry]*; ours (restated without the torsion hypothesis).*

Proof. The last sentence is [K:bp:p2:frame-basic](b)–(d), whose hypotheses are [K:bp:p2:A2]–[K:bp:p2:A4] (and [K:bp:p2:A6] for (c)); they contain no torsion condition.

Step S3.3: Derivatives of the error. E is holomorphic with $|E| \le \varepsilon$ on T_r ([K:bp:p2:frame-basic](a)); [K:bp:p1:cauchy-strip] gives $|DE| \le n\varepsilon/\delta$ on $T_{r-\delta}$.

Step S3.4: $\partial_\omega\Omega_K$ and E_{red}. By [K:bp:p2:frame-identities](d), (e), [K:bp:p2:matrix] and [K:bp:p2:frame-basic](b), as in [K:step:p2:geo-domegak], [K:step:p2:geo-ered]: $|\partial_\omega\Omega_K| \le (2n\mathfrak{m} + 2n^2\mathfrak{m})\varepsilon/\delta$ and $|E_{\mathrm{red}}| \le C_{\mathrm{red}}\varepsilon/\delta$ on $T_{r-\delta}$.

Step S3.5: Lagrangian defect and symplecticity defect. Ω_K is holomorphic, 1-periodic on $T_{r-\delta}$ with $\langle \Omega_K \rangle = 0$ ([K:bp:p2:exactness]; K primary). [K:bp:p2:cohomological](b) on $T_{r-\delta}$ with loss $\delta < r - \delta$ gives the bound of Ω_K on $T_{r-2\delta}$, and $|E_{\mathrm{sym}}| \le \mathfrak{m}^2|\Omega_K|$ by [K:bp:p2:frame-identities](b). □

Lemma 3.6 (Perturbation of the frame, without [K:bp:p2:A5])**.** *Let $r > 0$, $0 < \delta \le 1$, $2\delta < r$, $|\omega| \le \Theta$, $\omega \in \mathbb{R}^n$. Let K be holomorphic on T_r with $B_\beta(K(\theta)) \subset D$, $|DK| \le \sigma_K$, $\det G \neq 0$, $|B| \le \sigma_B$ on T_r. Let $\tilde{K}$ be holomorphic on T_r with $|\tilde{K} - K| \le d$ on T_r, where $d < \beta$ and $2\mathfrak{m}C_{\Delta T}\, d/\delta^2 \le 1$. Then (a)–(e) of* [K:bp:p2:perturbation] *hold:*

(a) $B_{\beta-d}(\tilde{K}(\theta)) \subset D$ on T_r and $|\tilde{A} - A| \le \mathfrak{m}d$;
(b) $|D\tilde{K} - DK| \le nd/\delta$ on $T_{r-\delta}$;
(c) $\det \tilde{G} \neq 0$, $|\tilde{B}| \le 2\mathfrak{m}$, $|\tilde{B} - B| \le C_B d/\delta$ on $T_{r-\delta}$;
(d) $|\tilde{N} - N| \le C_{\Delta N} d/\delta$, $|\tilde{N}| \le 2\mathfrak{m}^2$, $|\tilde{N}^\top| \le 4n\mathfrak{m}^2$ on $T_{r-\delta}$;
(e) $|\tilde{T} - T| \le C_{\Delta T} d/\delta^2$ on $T_{r-2\delta}$ (here $\tilde{T}$ is the torsion of $\tilde{K}$ for the same ω).

Source: HL19 Lemma 5.2, Steps 3–7, pp. 36–42, as in [K:bp:p2:perturbation]*; the torsion hypothesis there is used only for its item (f).*

Proof. **Step S3.6: Smallness.** From the definitions of [K:bp:p2:constants], $C_{\Delta T} \geq \max\{n, C_G, C_{\Delta N}\}$ ([K:step:p2:pt-small], pure arithmetic on constants). With $d/\delta \leq d/\delta^2 \leq 1/(2\mathfrak{m}C_{\Delta T})$: $nd/\delta \leq \mathfrak{m}$, $\mathfrak{m}C_G d/\delta \leq \frac{1}{2}$, $C_{\Delta N} d/\delta \leq \mathfrak{m}^2$, $\mathfrak{m}C_{\Delta T} d/\delta^2 \leq \frac{1}{2}$.

Step S3.7: Domain. $|z - \tilde{K}(\theta)| < \beta - d$ implies $|z - K(\theta)| < \beta$, so $z \in D$; the segment $[K, \tilde{K}]$ lies in $B_\beta(K(\theta)) \subset D$, and [K:bp:p2:hamiltonian](e) gives $|\tilde{A} - A| \leq Md \leq \mathfrak{m}d$.

Step S3.8: Frame. [K:bp:p1:cauchy-strip], [K:bp:p1:cauchy-corollaries] ($p = 2n$) on $\tilde{K} - K$: $|\tilde{L} - L| \leq nd/\delta$, $|(\tilde{L} - L)^\top| \leq 2nd/\delta$ on $T_{r-\delta}$; so $|\tilde{L}| \leq 2\mathfrak{m}$, $|\tilde{L}^\top| \leq 4n\mathfrak{m}$. Then $\tilde{G} - G = \tilde{L}^\top(\tilde{L} - L) + (\tilde{L} - L)^\top L$ has norm $\leq C_G d/\delta$, $|B||\tilde{G} - G| \leq \frac{1}{2}$, and [K:bp:p1:neumann] gives $\tilde{G}$ invertible, $|\tilde{B}| \leq 2\mathfrak{m}$, $|\tilde{B} - B| \leq C_B d/\delta$ (Gram invertibility before any use of $\tilde{N}$, $\tilde{T}$: Remark 1.13(T2)).

Step S3.9: Normal frame and torsion. $\tilde{N} - N = \Omega_0\big((\tilde{L} - L)\tilde{B} + L(\tilde{B} - B)\big)$ gives (d). On $T_{r-2\delta}$, $\tilde{X}_{\tilde{N}} - X_N = (\tilde{A} - A)\tilde{N} + A(\tilde{N} - N) - \partial_\omega(\tilde{N} - N)$, with the first two terms $\leq 2\mathfrak{m}^3 d + \mathfrak{m}C_{\Delta N} d/\delta$ and the third $\leq n\Theta C_{\Delta N} d/\delta^2$ ([K:bp:p1:cauchy-corollaries] with $|\omega|_1 \leq n\Theta$); so $|\tilde{X}_{\tilde{N}} - X_N| \leq C_{\Delta X} d/\delta^2$. Finally $\tilde{T} - T = \tilde{N}^\top \Omega_0 (\tilde{X}_{\tilde{N}} - X_N) + (\tilde{N} - N)^\top \Omega_0 X_N$ and $|X_N| \leq C_X/\delta$ (Lemma 3.5, last sentence) give $C_{\Delta T} d/\delta^2$. □

3.3. **The isoenergetic correction.**

Lemma 3.7 (Size of the isoenergetic correction)**.** *Let $s > 1$, $\omega = s\omega^*$, $0 < \delta \leq 1$, $2\delta < r$, $0 \leq \varepsilon \leq 1$, and let K be $(r, \beta, \varepsilon, s)$-admissible. Apply Lemma 2.12 with*

$$\eta^L = -N^\top \Omega_0 E, \quad \eta^N = L^\top \Omega_0 E, \quad \eta^\omega := -E^\omega(K), \quad T = T_{H,K,\omega}, \quad Y = DH(K)N, \quad \sigma = \sigma_T, \quad \xi_0^L := 0,$$

and put $\xi = (\xi^L, \xi^N)$, $\xi^\omega \in \mathbb{R}$, $\Delta K := P\xi$ on $T_{r-2\delta}$, $\Delta\omega := -\omega\xi^\omega$, $\bar{\omega} := \omega + \Delta\omega = \bar{s}\omega^*$ with $\bar{s} = s(1 - \xi^\omega)$. Then the lemma applies, $\xi, \Delta K$ are holomorphic, 1-periodic and real on reals, $\xi^\omega \in \mathbb{R}$, and with $\Gamma_\delta = c_R \gamma^{-1} \delta^{-\varkappa}$:

$$|\xi_0|, |\xi^\omega| \leq \frac{\hat{C}_S \varepsilon}{\gamma \delta^{\varkappa+1}}, \qquad |\xi^N| \leq \frac{\hat{C}_{\xi N} \varepsilon}{\gamma \delta^{\varkappa+1}}, \qquad |g^L| \leq \frac{\hat{C}_g \varepsilon}{\gamma \delta^{\varkappa+2}} \qquad \text{on } T_{r-\delta},$$

$$|\xi| \leq \frac{\hat{C}_\xi \varepsilon}{\gamma^2 \delta^a}, \quad |\Delta K| \leq \frac{\hat{C}_{\Delta K} \varepsilon}{\gamma^2 \delta^a}, \quad |\partial_\omega \xi| \leq \frac{\hat{C}_g \varepsilon}{\gamma \delta^{\varkappa+2}}, \quad |\partial_\omega \Delta K| \leq \frac{\hat{C}_\partial \varepsilon}{\gamma^2 \delta^{a+1}} \qquad \text{on } T_{r-2\delta},$$

$$|\Delta\omega| = |\bar{s} - s||\omega^*| \leq \frac{\Theta^2 \hat{C}_S \varepsilon}{\gamma^2 \delta^a}.$$

Consequently, with $d := \hat{C}_\Delta \varepsilon/(\gamma^2 \delta^a)$: $|\Delta K| \leq d$ on $T_{r-2\delta}$ and $|\Delta\omega| \leq d$.
Source: HL19 Lemma 6.1, pp. 46–47, Lemma 6.2 Step 1, pp. 48–49, (6.29)–(6.31), Table 4, p. 59.

Proof. **Step S3.10: Hypotheses of the border solve.** $\omega \in \mathcal{D}_{\gamma,\tau}$, $\gamma \leq |\omega| \leq \Theta$ by (2). By Lemma 3.5: $H_L = 2n\mathfrak{m}^2\varepsilon$, $H_N = 2n\mathfrak{m}\varepsilon$, $H_T = C_T/\delta$ on $T_{r-\delta}$; η^L, η^N, T are holomorphic, 1-periodic, real on reals ([K:bp:p2:periodicity], [K:bp:p2:frame-basic]); $\langle \eta^N \rangle = 0$ (Lemma 2.9(d)). Y is holomorphic, periodic, real on reals with $|Y| \leq M\sigma_K\sigma_B \leq \mathfrak{m}^3 =: H_Y$ (Lemma 2.5); $\mathcal{T} = \langle \mathsf{T}_H \rangle$ (same lemma(a)) is invertible with $|\mathcal{T}^{-1}| \leq \sigma_T \leq \mathfrak{m}$ by (B5); $\eta^\omega \in \mathbb{R}$ since $\langle H \circ K \rangle$ is real (Lemma 2.9(a)) and $E_0 \in \mathbb{R}$; $|\eta^\omega| \leq \varepsilon$ by (B7).

Step S3.11: The constants ξ_0, ξ^ω. In $S_* = \sigma \max\{H_L + H_T\Gamma_\delta H_N,\ |\eta^\omega| + H_Y \Gamma_\delta H_N\}$, using $\gamma\delta^j \leq \Theta$: first entry $\leq 2n\mathfrak{m}(\mathfrak{m}\Theta + c_R C_T)\varepsilon/(\gamma\delta^{\varkappa+1})$, so $\sigma\cdot$ it $\leq C_{\xi_0}\varepsilon/(\gamma\delta^{\varkappa+1})$; second entry $\leq (\Theta + 2nc_R\mathfrak{m}^4)\varepsilon/(\gamma\delta^\varkappa)$, so $\sigma\cdot$ it $\leq (\mathfrak{m}\Theta + 2nc_R\mathfrak{m}^5)\varepsilon/(\gamma\delta^{\varkappa+1})$. Hence $S_* \leq \hat{C}_S\varepsilon/(\gamma\delta^{\varkappa+1})$.

Step S3.12: ξ^N, g^L, ξ^L. Lemma 2.12(c): $|\xi^N| \le S_* + \Gamma_\delta H_N \le \hat{C}_{\xi N}\varepsilon/(\gamma\delta^{\varkappa+1})$; $|g^L| \le H_T(S_* + \Gamma_\delta H_N) + |\omega|S_* + H_L \le \hat{C}_g\varepsilon/(\gamma\delta^{\varkappa+2})$ (absorbing $\varepsilon \le \Theta\varepsilon/(\gamma\delta^{\varkappa+2})$); $|\xi^L| = |\xi^L - \xi_0^L| \le \Gamma_\delta \sup|g^L| = c_R\hat{C}_g\varepsilon/(\gamma^2\delta^{2\varkappa+2})$, and $|\xi^N| \le \Theta\hat{C}_{\xi N}\varepsilon/(\gamma^2\delta^a)$. So $|\xi| \le \hat{C}_\xi\varepsilon/(\gamma^2\delta^a)$ and $|\Delta K| \le |P||\xi| \le \hat{C}_{\Delta K}\varepsilon/(\gamma^2\delta^a)$. By Lemma 2.12(b), $\partial_\omega\xi = (g^L, -\eta^N)$ and $|\eta^N| \le 2n\mathfrak{m}\varepsilon \le \hat{C}_g\varepsilon/(\gamma\delta^{\varkappa+2})$.

Step S3.13: The frequency correction. $|\Delta\omega| = |\omega||\xi^\omega| \le \Theta\hat{C}_S\varepsilon/(\gamma\delta^{\varkappa+1}) \le \Theta^2\hat{C}_S\varepsilon/(\gamma^2\delta^a)$, and $\Delta\omega = -\xi^\omega s\,\omega^*$, so $|\Delta\omega| = |\bar{s} - s||\omega^*|$.

Step S3.14: Derivative of the correction along ω. $\partial_\omega L = AL - DE$ ([K:bp:p2:frame-identities](c)), so $|\partial_\omega L| \le \mathfrak{m}^2 + n\varepsilon/\delta$ on $T_{r-\delta}$; $\partial_\omega N = AN - X_N$, so $|\partial_\omega N| \le \mathfrak{m}^3 + C_X/\delta$. With $\delta, \varepsilon \le 1$: $|\partial_\omega P| \le C_P/\delta$ on $T_{r-\delta}$. Then $\partial_\omega\Delta K = (\partial_\omega P)\xi + P\partial_\omega\xi$ has norm $\le C_P\hat{C}_\xi\varepsilon/(\gamma^2\delta^{a+1}) + 2\mathfrak{m}^2\hat{C}_g\varepsilon/(\gamma\delta^{\varkappa+2}) \le \hat{C}_\partial\varepsilon/(\gamma^2\delta^{a+1})$ (as $\varkappa + 2 \le a + 1$).

Step S3.15: Regularity. By Lemma 2.12(a), (e) ξ is holomorphic, periodic, real on reals ($\xi_0^L = 0 \in \mathbb{R}^n$); P is too ([K:bp:p2:periodicity]); hence $\Delta K = P\xi$. The final sentence: $\hat{C}_{\Delta K}, \Theta^2\hat{C}_S \le \hat{C}_\Delta$. □

Lemma 3.8 (The new errors)**.** *In the setting of Lemma 3.7 assume moreover $d < \beta$ and $C_{\mathrm{sym}}\varepsilon/(\gamma\delta^{\varkappa+1}) \le \frac{1}{2}$. Put $\bar{K} := K + \Delta K$ on $T_{r-2\delta}$, $\bar{K} := K$ elsewhere. Then $\bar{K}(T_{r-2\delta}) \subset D$, $S := I_{2n} - \Omega_0 E_{\mathrm{sym}}$ is invertible with $|S^{-1}| \le 2$ on $T_{r-2\delta}$, and on $T_{r-2\delta}$*

$$\bar{E} := E_{H,\bar{K},\bar{\omega}} = PS^{-1}E_{\mathrm{lin}} + \Delta^2 X + \xi^\omega\,\partial_\omega\Delta K, \tag{3}$$

$$E_{\mathrm{lin}} := E_{\mathrm{red}}\xi + \Omega_0 E_{\mathrm{sym}}\,\partial_\omega\xi - \begin{pmatrix} 0 \\ \Omega_K\omega \end{pmatrix}\xi^\omega, \qquad \Delta^2 X := X_H(\bar{K}) - X_H(K) - A\Delta K.$$

Moreover $\bar{E}^\omega := \bar{E}_H(\bar{K}) - E_0 = E^\omega_{\mathrm{lin}} + \langle\Delta^2 H\rangle$ with $E^\omega_{\mathrm{lin}} = \langle D(H\circ K)\xi^L\rangle$, $\Delta^2 H := H(\bar{K}) - H(K) - DH(K)\Delta K$, and

$$|\bar{E}| \le \frac{\hat{C}_E\,\varepsilon^2}{\gamma^4\delta^{2a}} \quad \text{on } T_{r-2\delta}, \qquad |\bar{E}^\omega| \le \frac{\hat{C}_E\,\varepsilon^2}{\gamma^4\delta^{2a}}.$$

Source: HL19 Lemma 6.2 Step 2, p. 49–50, (6.34)–(6.39); $L_{\Delta\omega}\Delta K = -\xi^\omega L_\omega\Delta K$ is the last term of (3).

Proof. **Step S3.16: Invertibility and domain.** As in [K:step:p2:ne-invert], [K:step:p2:ne-domain]: $|\Omega_0 E_{\mathrm{sym}}| \le \frac{1}{2}$ (Lemma 3.5), [K:bp:p1:neumann] gives $|S^{-1}| \le 2$; $QP = S$ ([K:bp:p2:frame-identities](f)) so $Q^{-1} = PS^{-1}$. $|\Delta K| \le d < \beta$ puts $\bar{K}(\theta)$ in $B_\beta(K(\theta)) \subset D$, and the segment $[K, \bar{K}]$ too.

Step S3.17: Expansion. On $T_{r-2\delta}$: $\bar{E} = X_H(K + \Delta K) - (L + D\Delta K)(\omega + \Delta\omega) = E + A\Delta K - \partial_\omega\Delta K - L\Delta\omega + \Delta^2 X - D\Delta K\,\Delta\omega$, and $D\Delta K\,\Delta\omega = -\xi^\omega\partial_\omega\Delta K$.

Step S3.18: Reduction by Q. By Lemma 2.14(b), $Q(E + A\Delta K - \partial_\omega\Delta K - L\Delta\omega) = \Lambda\xi - \partial_\omega\xi + \binom{\omega}{0}\xi^\omega - \eta + E_{\mathrm{lin}}$, which equals E_{lin} by the first two equations of Lemma 2.12(b) (the data are those of the correction). Apply $Q^{-1} = PS^{-1}$; this is (3).

Step S3.19: Estimates. By Lemmas 3.5, 3.7 on $T_{r-2\delta}$: $|E_{\mathrm{red}}\xi| \le C_{\mathrm{red}}\hat{C}_\xi\varepsilon^2/(\gamma^2\delta^{a+1})$; $|\Omega_0 E_{\mathrm{sym}}\partial_\omega\xi| \le C_{\mathrm{sym}}\hat{C}_g\varepsilon^2/(\gamma^2\delta^{2\varkappa+3})$ with $2\varkappa + 3 = a + 1$; $|\Omega_K\omega\xi^\omega| \le \Theta C_\Omega\hat{C}_S\varepsilon^2/(\gamma^2\delta^{2\varkappa+2}) \le$ the same with δ^{a+1}. So $|E_{\mathrm{lin}}| \le \hat{C}_{\mathrm{lin}}\varepsilon^2/(\gamma^2\delta^{a+1})$. By [K:bp:p2:hamiltonian](e), $|\Delta^2 X| \le M|\Delta K|^2 \le \mathfrak{m}\hat{C}^2_{\Delta K}\varepsilon^2/(\gamma^4\delta^{2a})$. Also $|\xi^\omega\partial_\omega\Delta K| \le \hat{C}_S\hat{C}_\partial\varepsilon^2/(\gamma^3\delta^{\varkappa+a+2})$. Using $\gamma^2\delta^{a-1} \le \Theta^2$ and $\gamma\delta^{a-\varkappa-2} \le \Theta$ ($a - \varkappa - 2 = \varkappa \ge 0$) to put everything over $\gamma^4\delta^{2a}$, and $|P||S^{-1}| \le 4\mathfrak{m}^2$: $|\bar{E}| \le \hat{C}_{E1}\varepsilon^2/(\gamma^4\delta^{2a})$.

Step S3.20: The energy error. $\bar{K}$ is real on reals, continuous on $\mathbb{R}^n \subset T_{r-2\delta}$, so $H\circ\bar{K}$ is integrable. Pointwise on $\mathbb{R}^n$: $H(\bar{K}) = H(K) + DH(K)\Delta K + \Delta^2 H$, so $\bar{E}^\omega = E^\omega + \langle DH(K)\Delta K\rangle + \langle\Delta^2 H\rangle$. By Lemma 2.9(f) and the third equation of the border system, $\langle DH(K)\Delta K\rangle = \eta^\omega + E^\omega_{\mathrm{lin}} = -E^\omega + E^\omega_{\mathrm{lin}}$. By [K:bp:p1:taylor] ($F = H$ on $U = D$, $|D^2 H| \le M$, segment in D): $|\Delta^2 H| \le \frac{M}{2}|\Delta K|^2 \le \mathfrak{m}\hat{C}^2_{\Delta K}\varepsilon^2/(\gamma^4\delta^{2a})$. By Lemma 2.9(f), on $\mathbb{R}^n$: $|D(H\circ K)| \le n|\eta^N| + n|\omega||\Omega_K| \le$

$n\Theta(2n\mathfrak{m}+C_\Omega)\varepsilon/(\gamma\delta^{\varkappa+1})$, and $|\xi^L| \le c_R\hat{C}_g\varepsilon/(\gamma^2\delta^a)$; the product is $\le \hat{C}_{E2}\varepsilon^2/(\gamma^4\delta^{2a})$ (using $\gamma\delta^{\varkappa+1} \le \Theta$). □

Remark 3.9 (Why [K:bp:p2:newton-error] does not apply). Its hypothesis is that $\Delta K = P\xi$ with ξ from [K:bp:p2:triangular], i.e. $\Lambda\xi - \partial_\omega\xi = \eta$. Here $\Lambda\xi - \partial_\omega\xi = \eta - \binom{\omega}{0}\xi^\omega$. For the frozen frequency the first bracket is $X_H(K+\Delta K) - D(K+\Delta K)\omega = PS^{-1}E_{\rm lin} + L\Delta\omega + \Delta^2 X$ (Step S3.18), which carries the *first-order* term $L\Delta\omega = -L\omega\xi^\omega = O(\varepsilon)$, cancelled only by the frequency change. The variant therefore has the new frequency in the error, the extra defect $-D\Delta K\,\Delta\omega$, and the extra term $\Omega_K\omega\xi^\omega$ in $E_{\rm lin}$.

3.4. **The new extended torsion and the step.**

Lemma 3.10 (The new averaged extended torsion)**.** *Let K be (r,β,ε,s)-admissible, $0<\delta\le 1$, $4\delta<r$, and let $\bar{K}$, $\bar{\omega}=\bar{s}\omega^*$ be as in Lemma 3.8. Then $\bar{\omega}\in\mathbb{R}^n$ and T_H for $(\bar{K},\bar{\omega})$ is defined on $T_{r-3\delta}$ (no Diophantine property of $\bar{\omega}$ is used here). Assume $d<\beta$ and $2\mathfrak{m}C_{\Delta\mathsf{T}}\,d/\delta^2\le 1$. Then $\bar{K}$ satisfies (B1)–(B4) with the bounds $B_{\beta-d}(\bar{K})\subset D$ on $T_{r-2\delta}$, $|D\bar{K}-DK|\le nd/\delta$ and $|\bar{B}-B|\le C_B d/\delta$ on $T_{r-3\delta}$; the matrix $\langle\bar{\mathsf{T}}\rangle := \langle\mathsf{T}_H(\bar{K},\bar{\omega})\rangle$ is defined, invertible, and*

$$|\langle\bar{\mathsf{T}}\rangle - \langle\mathsf{T}\rangle| \le C_{\Delta\mathsf{T}}\frac{d}{\delta^2}, \qquad |\langle\bar{\mathsf{T}}\rangle^{-1} - \langle\mathsf{T}\rangle^{-1}| \le 2\mathfrak{m}^2 C_{\Delta\mathsf{T}}\frac{d}{\delta^2}.$$

Source: HL19 Lemma 6.2 Step 7, p. 51, (6.43)–(6.44); Lemma A.1, p. 55.

Proof. **Step S3.21: Frame.** Apply Lemma 3.6 on $T_{r-2\delta}$ (so $2\delta<r-2\delta$) with $\tilde{K}=\bar{K}$, $|\bar{K}-K|\le d$ on $T_{r-2\delta}$ (Lemma 3.7); the smallness holds as $C_{\Delta\mathsf{T}}\ge C_{\Delta T}$. This gives the stated ball, derivative and Gram-inverse bounds, $|\bar{N}-N|\le C_{\Delta N}d/\delta$, $|\bar{N}|\le 2\mathfrak{m}^2$ on $T_{r-3\delta}$ and $|\bar{T}_\omega - T_\omega|\le C_{\Delta T}d/\delta^2$ on $T_{r-4\delta}$; $\bar{N}$, $\bar{T}_\omega$ are holomorphic on $T_{r-3\delta}\supset\mathbb{R}^n$, so $\mathsf{T}_H(\bar{K},\bar{\omega})$ is continuous on $\mathbb{R}^n$ and its average is defined (Lemma 2.5, applied to $\bar{K}$ on $T_{r-3\delta}$).

Step S3.22: Effect of the frequency on the torsion block. The torsion entries are $\bar{N}_i\cdot\Omega_0(\bar{A}\bar{N}_j - D\bar{N}_j\,\omega')$ with $\omega'=\omega$ or $\bar{\omega}$; so $T_{\bar{\omega}}(\bar{K}) - T_\omega(\bar{K}) = -\bar{N}^\top\Omega_0\,[D\bar{N}_j\Delta\omega]_j$. By [K:bp:p1:cauchy-corollaries] on $T_{r-3\delta}$ with loss δ, $|\partial_{\Delta\omega}\bar{N}|\le n|\Delta\omega|\,2\mathfrak{m}^2/\delta$ on $T_{r-4\delta}$, and $|\bar{N}^\top|\le 4n\mathfrak{m}^2$; hence the difference has norm $\le 8n^2\mathfrak{m}^4 d/\delta$ on $T_{r-4\delta}$ (as $|\Delta\omega|\le d$).

Step S3.23: Effect on the energy row. $\bar{Y}-Y = (DH(\bar{K})-DH(K))\bar{N} + DH(K)(\bar{N}-N)$. By [K:bp:p1:increments-open] applied to $F=DH$ ($|DF|=|D^2H|\le M$, segment in D), $|DH(\bar{K})-DH(K)|\le Md$. Each entry satisfies $|\bar{Y}_b - Y_b|\le Md\,2\mathfrak{m}^2 + M\,C_{\Delta N}d/\delta$, and the row-sum norm of the $1\times n$ row is at most n times this: $|\bar{Y}-Y|\le C_Y d/\delta$ on $T_{r-3\delta}$ (here $\delta\le 1$).

Step S3.24: The average. The difference $\langle\bar{\mathsf{T}}\rangle - \langle\mathsf{T}\rangle$ is the average ([K:bp:p2:averages](a)) of $\begin{pmatrix}\bar{T}_{\bar{\omega}}-T_\omega & \Delta\omega\\ \bar{Y}-Y & 0\end{pmatrix}$ on $\mathbb{R}^n\subset T_{r-4\delta}$; by [K:bp:p2:matrix](c) its norm is $\le \max\{C_{\Delta T}d/\delta^2 + 8n^2\mathfrak{m}^4 d/\delta + d,\ C_Y d/\delta\} \le C_{\Delta\mathsf{T}}d/\delta^2$.

Step S3.25: Inverse (Lemma A.1). $|\langle\mathsf{T}\rangle^{-1}|\le\sigma_T\le\mathfrak{m}$, so $|\langle\mathsf{T}\rangle^{-1}||\langle\bar{\mathsf{T}}\rangle - \langle\mathsf{T}\rangle|\le\frac12$, and [K:bp:p1:neumann] (the max-row-sum norm of $\mathbb{C}^{(n+1)\times(n+1)}$; HL19 Lemma A.1) gives invertibility and the difference bound $2\sigma_T^2\le 2\mathfrak{m}^2$. □

Proposition 3.11 (The isoenergetic step)**.** *Let $r>0$, $0<\delta\le 1$ with $4\delta<r$, $\beta>0$, $0\le\varepsilon\le 1$, $s>1$ and let K be (r,β,ε,s)-admissible with*

$$\hat{\mathfrak{c}}_{\rm step}\frac{\varepsilon}{\gamma^4\delta^{2a}} \le \min\{1,\beta\}.$$

Let $\bar{K},\bar{\omega}=\bar{s}\omega^*$ be as in Lemmas 3.7, 3.8, and $d := \hat{C}_\Delta\varepsilon/(\gamma^2\delta^a)$. Then

(a) $\bar{K}$ is holomorphic, primary on $T_{r-2\delta}$, real on reals;
(b) $|\bar{K}-K|\le d\le\beta/2$ on $T_{r-2\delta}$, $B_{\beta-d}(\bar{K}(\theta))\subset D$ there, and $|\bar{s}-s||\omega^*|\le d$;

(c) $|E_{H,\bar K,\bar\omega}| \le \hat C_E\varepsilon^2/(\gamma^4\delta^{2a})$ *on* $T_{r-2\delta}$ *and* $|\bar E_H(\bar K)-E_0| \le \hat C_E\varepsilon^2/(\gamma^4\delta^{2a})$*;*
(d) $|D\bar K - DK| \le nd/\delta$ *and* $\det \bar G \neq 0$*,* $|\bar B - B| \le C_B d/\delta$ *on* $T_{r-3\delta}$*;*
(e) $\langle \mathsf{T}_H(\bar K,\bar\omega)\rangle$ *is invertible and* $|\langle\bar{\mathsf{T}}\rangle^{-1} - \langle\mathsf{T}\rangle^{-1}| \le 2\mathfrak{m}^2 C_{\Delta\mathsf{T}} d/\delta^2$*.*

Source: HL19 Lemma 6.2, pp. 47–51.

Proof. **Step S3.26: Smallness.** Put $X := \varepsilon/(\gamma^4\delta^{2a}) \le \min\{1,\beta\}/\hat{\mathfrak{c}}_{\mathrm{step}}$. With $\gamma \le \Theta$ (2) and $\delta \le 1$: $C_{\mathrm{sym}}\varepsilon/(\gamma\delta^{\varkappa+1}) \le \Theta^3 C_{\mathrm{sym}} X \le \frac12$; $d \le \Theta^2\hat C_\Delta X \le \beta/2$; $2\mathfrak{m}C_{\Delta\mathsf{T}}d/\delta^2 \le 2\mathfrak{m}\Theta^2 C_{\Delta\mathsf{T}}\hat C_\Delta X \le 1$ (as $\gamma^2\delta^{a-2} \le \Theta^2$).

Step S3.27: Correction and errors. Lemmas 3.7 (with $2\delta < r$) and 3.8 give (a) ($\bar K(\theta+k) = K(\theta) + (k,0) + \Delta K(\theta)$ by periodicity of ΔK), (b) except the ball, and (c) (note $\hat C_E$ bounds both errors).

Step S3.28: Ball, frame, torsion. The ball is Lemma 3.6(a); (d), (e) are Lemma 3.10. □

3.5. Schedule, decay, induction and limit.

Definition 3.12 (Schedule)**.** Given $0 < \rho_\infty < \rho$, $\gamma > 0$, $e \ge 0$: $\delta_s := 2^{-s}\delta_0$, $r_s := \rho - 8\delta_0(1-2^{-s})$, $\varepsilon_0 := e$, $\varepsilon_{s+1} := \hat C_E\varepsilon_s^2/(\gamma^4\delta_s^{2a})$, $d_s := \hat C_\Delta\varepsilon_s/(\gamma^2\delta_s^a)$. Then $r_0 = \rho$, $r_{s+1} = r_s - 4\delta_s$, $\rho_\infty \le \rho - 8\delta_0 < r_s$, $\delta_s \le 1$, $4\delta_s < r_s$ (as [K:`bp:p2:schedule`]).
Source: HL19 §6.3, p. 52.

Lemma 3.13 (Decay)**.** *If* $q := 2^{2a}\hat C_E e/(\gamma^4\delta_0^{2a}) \le 1$*, then* $\varepsilon_s \le q^{2^s-1}2^{-2as}e \le 2^{-2as}e$*; hence* $\varepsilon_s/\delta_s^j \le e/\delta_0^j$ *for* $0 \le j \le 2a$ *and* $\sum_s \varepsilon_s/\delta_s^j \le 2e/\delta_0^j$ *for* $0 \le j \le 2a-1$*.*

Proof. Identical to [K:`bp:p2:error-decay`] (it uses only $\varepsilon_{s+1} = Q'2^{2as}\varepsilon_s^2$ with $Q' = \hat C_E/(\gamma^4\delta_0^{2a})$ and [K:`bp:p1:iteration`] with $p = 2a$; the geometric series as there). □

Proposition 3.14 (The induction)**.** *Let* K*,* $s = s_0 > 1$ *satisfy (I5)–(I12) of Theorem 1.14 on* T_ρ *with the margin* μ*: (B1);* $B_\mu(K) \subset D$*;* $|DK| \le \sigma_K - \mu$*;* $\det G \neq 0$*,* $|B| \le \sigma_B - \mu$*;* $\det\langle\mathsf{T}_H\rangle \neq 0$*,* $|\langle\mathsf{T}_H\rangle^{-1}| \le \sigma_T - \mu$*;* $|E_{H,K,s\omega^*}| \le e$*,* $|\bar E_H(K) - E_0| \le e$*; and* $\nu(s_0) \ge \nu$ *(I9). Assume*

$$e \le 1, \qquad \hat{\mathfrak{c}}_* \frac{e}{\gamma^4\delta_0^{2a}} \le 1.$$

Then there are K_j*,* $s_j \in \mathbb{R}$ *(*$K_0 = K$*) such that with* $D_j := \sum_{i<j} d_i$*, for every* j*:*

(i) K_j *is* $(r_j, \mu - D_j, \varepsilon_j, s_j)$*-admissible and* $\varepsilon_j \le e \le 1$*;*
(ii) $D_j \le \mu/4$ *and* $D_j \le \nu/2$*;* $|K_j - K| \le D_j$ *on* T_{r_j}*;* $|s_j - s_0||\omega^*| \le D_j$ *and* $\nu(s_j) \ge \nu - D_j \ge \nu/2$*, so* $1 < s_j < \sigma_\omega$ *(Lemma 3.2);*
(iii) $|DK_j| \le \sigma_K - \mu + \sum_{i<j} nd_i/\delta_i \le \sigma_K - \mu/2$ *on* T_{r_j}*;*
(iv) $|B_j| \le \sigma_B - \mu + \sum_{i<j} C_B d_i/\delta_i \le \sigma_B - \mu/2$ *on* T_{r_j}*;*
(v) $|\langle\mathsf{T}_j\rangle^{-1}| \le \sigma_T - \mu + \sum_{i<j} 2\mathfrak{m}^2 C_{\Delta\mathsf{T}} d_i/\delta_i^2 \le \sigma_T - \mu/2$*,* $\mathsf{T}_j := \mathsf{T}_H(K_j, s_j\omega^*)$*;*
(vi) $|K_{j+1} - K_j| \le d_j$*,* $|DK_{j+1} - DK_j| \le nd_j/\delta_j$ *on* $T_{r_{j+1}}$*,* $|s_{j+1} - s_j||\omega^*| \le d_j$*.*

Source: HL19 §6.3, p. 52 (the eighth term of (6.15) is checked there as here, via the margin).

Proof. **Step S3.29: Summed bounds.** K is (ρ,μ,e,s_0)-admissible, so $\gamma \le \Theta$ by (2). Put $X := e/(\gamma^4\delta_0^{2a}) \le 1/\hat{\mathfrak{c}}_*$. Since $e/(\gamma^2\delta_0^i) \le \Theta^2 X$ for $i \le 2a$, Lemma 3.13 ($\hat{\mathfrak{c}}_* \ge 2^{2a}\hat C_E$; $a+2 \le 2a-1$ as $a \ge 8$) gives $\sum_i d_i/\delta_i^k \le 2\Theta^2\hat C_\Delta X$ for $k = 0,1,2$. By the choice of $\hat{\mathfrak{c}}_*$: $\sum d_i \le \min\{\mu/4, \nu/2\}$, $\sum nd_i/\delta_i \le \mu/4$, $\sum C_B d_i/\delta_i \le \mu/4$, $\sum 2\mathfrak{m}^2 C_{\Delta\mathsf{T}} d_i/\delta_i^2 \le \mu/2$ (these use $\hat{\mathfrak{c}}_* \ge 2\hat{\mathfrak{c}}_{\mathrm{sum}}/\mu$ and $\ge 4\Theta^2\hat C_\Delta/\nu$).

Step S3.30: Step smallness. $\varepsilon_j \le e \le 1$, $\varepsilon_j/\delta_j^{2a} \le e/\delta_0^{2a}$, so $\hat{\mathfrak{c}}_{\mathrm{step}}\varepsilon_j/(\gamma^4\delta_j^{2a}) \le \hat{\mathfrak{c}}_{\mathrm{step}} X \le \min\{1, \mu/2\} \le \min\{1, \mu - D_j\}$.

Step S3.31: Base and induction. $j = 0$: $K_0 = K$ is (ρ,μ,e,s_0)-admissible; (ii)–(v) hold with empty sums. Assume (i)–(v) at j. Proposition 3.11 applies (with $r_j, \delta_j, \mu - D_j, \varepsilon_j, s_j$) and yields

$\bar K_j$, $\bar s_j$ with $|\bar s_j - s_j||\omega^*| \le d_j$. Since $d_j \le D_{j+1} - D_j \le \nu/2 - D_j < \nu - D_j \le \nu(s_j)$ (Step S3.29), Lemma 3.2(b) gives $1 < \bar s_j < \sigma_\omega$ and $\nu(\bar s_j) \ge \nu(s_j) - d_j \ge \nu - D_{j+1}$. Set $K_{j+1} := \bar K_j$, $s_{j+1} := \bar s_j$. By (b)–(e) on $T_{r_{j+1}} \subset T_{r_j - 3\delta_j}$: (B1); $B_{\mu - D_{j+1}}(K_{j+1}) \subset D$; $|E_{j+1}|$, $|E^\omega_{j+1}| \le \varepsilon_{j+1}$ (B6, B7); (vi); $|K_{j+1} - K| \le D_j + d_j$; $|DK_{j+1}| \le |DK_j| + n d_j/\delta_j$; $|B_{j+1}| \le |B_j| + C_B d_j/\delta_j$; $\det\langle \mathsf{T}_{j+1}\rangle \ne 0$ with the stated inverse bound; and by Lemma 3.2(b), $\nu(s_{j+1}) \ge \nu(s_j) - d_j \ge \nu - D_{j+1}$. With the summed bounds this is (i)–(v) at $j+1$ ((B3)–(B5) follow as in [K:`step:p2:it-step`]). □

Proposition 3.15 (The limit)**.** *In the setting of Proposition 3.14 put $K_\infty := \lim_j K_j$ on T_{ρ_∞}, $K_\infty := K$ elsewhere, and $s_\infty := \lim_j s_j$. Then:*

- *(a) the limits exist; $K_j \to K_\infty$ and $DK_j \to DK_\infty$ uniformly on T_{ρ_∞}; K_∞ is holomorphic on T_{ρ_∞}, primary there and real on reals;*
- *(b) $|K_\infty - K| \le 2\hat C_\Delta e/(\gamma^2\delta_0^a) \le \mu/4$ on T_{ρ_∞}, and $|s_\infty\omega^* - s_0\omega^*| \le 2\hat C_\Delta e/(\gamma^2\delta_0^a) \le \nu/2$;*
- *(c) $1 < s_\infty < \sigma_\omega$ (indeed $\nu(s_\infty) \ge \nu/2$);*
- *(d) $B_{\mu/2}(K_\infty(\theta)) \subset D$ and $|DK_\infty(\theta)| \le \sigma_K - \mu/2$ on T_{ρ_∞};*
- *(e) $E_{H,K_\infty,s_\infty\omega^*} = 0$ on T_{ρ_∞}.*

Source: HL19 p. 52–53, (6.50)–(6.51).

Proof. **Step S3.32: Convergence.** $T_{\rho_\infty} \subset T_{r_j}$; by (vi) $\sum_j d_j$, $\sum_j n d_j/\delta_j$ converge ($\le 2\hat C_\Delta e/(\gamma^2\delta_0^a)$, Lemma 3.13), so K_j, DK_j are uniformly Cauchy and (s_j) is Cauchy ($\omega^* \ne 0$). [K:`bp:p1:uniform-limits`] gives holomorphy, primary, real, and $DK_\infty = \lim DK_j$ (this is (a)).

Step S3.33: Closeness and frequency. $|K_\infty - K| \le \sup_j D_j$ and $|s_\infty - s_0||\omega^*| \le \sup_j D_j \le \min\{\mu/4, \nu/2\}$ with the bound of Step S3.29: (b). Passing to the limit in $\nu(s_j) \ge \nu - D_j$ (continuity of $\nu(\cdot)$) gives $\nu(s_\infty) \ge \nu/2 > 0$: (c).

Step S3.34: Margins. If $|z - K_\infty(\theta)| < \mu/2$, then $|z - K(\theta)| < 3\mu/4$, so $z \in D$; $|DK_\infty| = \lim |DK_j| \le \sigma_K - \mu/2$ by (iii): (d).

Step S3.35: Invariance. For $\theta \in T_{\rho_\infty}$, $K_\infty(\theta) \in D$ where X_H is continuous, so $E_j(\theta) = X_H(K_j(\theta)) - DK_j(\theta) s_j \omega^* \to E_{H,K_\infty,s_\infty\omega^*}(\theta)$; and $|E_j| \le \varepsilon_j \le 2^{-2aj} e \to 0$. □

Lemma 3.16 (Passage of the energy average to the limit)**.** *In the setting of Proposition 3.15, $\bar E_H(K_\infty) = E_0$, and $\bar E_H(K_j) \to \bar E_H(K_\infty)$.*

Proof. **Step S3.36: Uniform convergence of $H \circ K_j$ on $\mathbb{R}^n$.** For $\theta \in \mathbb{R}^n$ both $K_j(\theta)$ and $K_\infty(\theta)$ lie in $B_\mu(K(\theta)) \subset D$ (distance to $K(\theta)$ at most $\mu/4$), a convex subset of D. By [K:`bp:p1:increments-open`] ($F = H$, $|DH| \le M$): $|H(K_j) - H(K_\infty)| \le M|K_j - K_\infty| \to 0$ uniformly on $\mathbb{R}^n$.

Step S3.37: Integrals. $H \circ K_\infty$ is continuous on $\mathbb{R}^n$, 1-periodic (K primary, H periodic), so integrable on $[0,1]^n$ ([K:`bp:p1:ext-integral`]); $|\bar E_H(K_j) - \bar E_H(K_\infty)| \le \sup_{\mathbb{R}^n} |H(K_j) - H(K_\infty)| \to 0$ (volume one). Since $|\bar E_H(K_j) - E_0| \le \varepsilon_j \to 0$ (B7), $\bar E_H(K_\infty) = E_0$. □

3.6. **Proof of I-G1.**

Theorem 3.17 (I-G1 with explicit constants)**.** *Let the data of Theorem 1.14 be fixed $(n, \tau, \rho, \rho_\infty, M, \sigma_K, \sigma_B, \sigma_T, \mu, \bar\gamma, \sigma_\omega, \nu, \omega^*)$. With Definition 3.3 put*

$$C_1 := \rho^{4\tau} \max\{\bar\gamma^4,\ \hat{\mathfrak{c}}_* \delta_0^{-2a}\}, \qquad C_2 = C_3 := 2\hat C_\Delta\, \rho^{2\tau} \delta_0^{-a}.$$

Then $C_1, C_2, C_3 > 0$ depend only on $(n, \tau, \rho, \rho_\infty, M, \sigma_K, \sigma_B, \sigma_T, \mu, \bar\gamma, \nu)$ (in particular they are fixed before $D, H, K, s, \gamma, e, E_0$, and do not use ω^*, σ_ω), and the conclusions (O1)–(O6) of Theorem 1.14 hold for all $D, H, K, s, \gamma, e, E_0$ satisfying (I1)–(I13).

Frozen Lean declaration: `KAM.Iso.isoenergetic_kam`

Source: HL19 Theorem 3.6 and Rem. 3.7, pp. 18–19; proof pp. 45–53; Table 5, p. 59, (6.45)–(6.51).

Proof. **Step S3.38: The constants.** All constants of Definition 3.3 are finite, positive, and $\delta_0 \in (0,1]$, $\rho > 0$; so $C_1 \geq \rho^{4\tau}\bar{\gamma}^4 > 0$ and $C_2 > 0$. Their dependence is read off the definitions; $\rho^{4\tau}, \rho^{2\tau}$ are real powers of a positive number.

Step S3.39: Standing data. (I1)–(I4) and $E_0 \in \mathbb{R}$, $\gamma > 0$, $\omega^* \in \mathcal{D}_{\gamma,\tau}$ (I10) are Assumption 3.1.

Step S3.40: Size of the errors. From (I13) and $C_1 \geq \rho^{4\tau}\bar{\gamma}^4$: $e < \gamma^4/\bar{\gamma}^4 \leq 1$ ($\gamma \leq \bar{\gamma}$, I10); and $\hat{\mathfrak{c}}_* e/(\gamma^4\delta_0^{2a}) < 1$ from $C_1 \geq \rho^{4\tau}\hat{\mathfrak{c}}_*\delta_0^{-2a}$.

Step S3.41: Dictionary for the hypotheses. (I5): (B1). (I6): $B_\mu(K) \subset D$, $|DK| \leq \sigma_K - \mu$. (I7): $\det G \neq 0$, $|B| \leq \sigma_B - \mu$ (Gram invertibility is a hypothesis, so N, T are genuine, Remark 1.13(T2)). (I8): $\det\langle \mathsf{T}_H\rangle \neq 0$, $|\langle \mathsf{T}_H\rangle^{-1}| \leq \sigma_T - \mu$; the average is defined by Lemma 2.5. (I9): $s > 1$ and $\nu(s) \geq \nu$. (I11): (B6). (I12): (B7). $s > 1$ gives $s\omega^* \in \mathcal{D}_{\gamma,\tau}$ (Lemma 3.2).

Step S3.42: Iteration and limit. Propositions 3.14, 3.15 apply and give K_∞, s_∞.

Step S3.43: (O1), (O4)–(O6). (O1) is Proposition 3.15(a); (O4) is (d). (O5): $|K_\infty - K| \leq 2\hat{C}_\Delta e/(\gamma^2\delta_0^a) = C_2 e/(\gamma^2\rho^{2\tau})$ on T_{ρ_∞}. (O6): $|s_\infty\omega^* - s\omega^*| \leq 2\hat{C}_\Delta e/(\gamma^2\delta_0^a) = C_3 e/(\gamma^2\rho^{2\tau})$ (b). Also $1 < s_\infty < \sigma_\omega$ by (c).

Step S3.44: (O2). Proposition 3.15(e).

Step S3.45: (O3). K_∞ is holomorphic, primary, real on reals on T_{ρ_∞} with $K_\infty(T_{\rho_\infty}) \subset D$ (by (d)), so Assumption 2.1 holds with $r = \rho_\infty$, $\omega = s_\infty\omega^* \in \mathcal{D}_{\gamma,\tau}$; $E_{H,K_\infty,s_\infty\omega^*} = 0$ there (O2) and $\bar{E}_H(K_\infty) = E_0$ (Lemma 3.16). Lemma 2.10(b) gives $H(K_\infty(\theta)) = E_0$ for all $\theta \in T_{\rho_\infty}$ (Remark 2.11); no frame or Gram determinant of K_∞ is used. □

Remark 3.18 (Register of constants). $\hat{C}_S, \hat{C}_{\xi N}, \hat{C}_g, \hat{C}_\xi, \hat{C}_{\Delta K}, \hat{C}_\Delta$: Lemma 3.7 (HL19 $C_{\xi_0^N}, C_{\xi^\omega}, C_\omega, C_{\Delta\omega}$, (6.29)–(6.31)); $C_P, \hat{C}_\partial, \hat{C}_{\mathrm{lin}}, \hat{C}_{E1}, \hat{C}_{E2}, \hat{C}_E$: Lemma 3.8 ($C_{\mathrm{lin}}, C^\omega_{\mathrm{lin}}, C_{L\Delta K}, C_E, C_{E^\omega}, C_{E_c}$, (6.34)–(6.39)); $C_Y, C_{\Delta\mathsf{T}}$: Lemma 3.10 ($C_{\Delta T_c}$, (6.43)); $\hat{\mathfrak{c}}_{\mathrm{step}}$: Proposition 3.11; $\hat{\mathfrak{c}}_{\mathrm{sum}}, \hat{\mathfrak{c}}_*$: Proposition 3.14 ((6.46)–(6.49)); C_1, C_2, C_3: Theorem 3.17 ((6.45), (6.50), (6.51)).

Remark 3.19 (Deviations from HL19 and from track K). (W1) As in track K: open strips with pointwise bounds, the exponent $\varkappa = \tau + n$, γ-independent step constants ($\gamma \leq |\omega| \leq \Theta$ from [K:`bp:p2:frequency`]), uniform margin μ, loss $4\delta_s$ per step; HL19's ν of (4.40)/(6.28) is replaced by $e \leq 1$, $\delta \leq 1$; the frozen ν is HL19's $\mathrm{dist}(\omega, \partial\Theta)$ (relative boundary: the two endpoints, review finding I01). (W2) $|\omega| \leq \Theta$ comes from [K:`bp:p2:frequency`] at every iterate, not from $|\omega| < \sigma_\omega|\omega^*|$ (HL19 (6.30)); the latter is unused, which is why C_i do not depend on ω^*, σ_ω. (W3) The first-order term $L\Delta\omega$ and the term $-\binom{0}{\Omega_K\omega}\xi^\omega$ are carried explicitly (Remark 3.9); HL19 p. 49 print E_{lin} with the signs of their $\mathfrak{L}_\omega = -\partial_\omega$. (W4) Conclusions (O5), (O6) read $\leq$ (not $<$), (O3) is pointwise (Part 2); (O6) is in the form $C_3 e/(\gamma^2\rho^{2\tau})$ of the frozen statement.

Remark 3.20 (Candidate errata and doubts). (E4) HL19 (6.51), p. 53: "$|\omega_\infty - \omega| < C_3\|E_c\|_\rho/(\gamma\rho^\tau)$", with $C_3 = a_3^\tau C_{\Delta\omega}/(1 - a_1^{-3\tau})$ (Table 5), then "thus obtaining the estimates in (3.16)", where (3.16) has $\gamma^2\rho^{2\tau}$. The exponents of γ and ρ differ, so (6.51) does not literally give (3.16). The proof here obtains the form of (3.16) directly (d_j has $\gamma^{-2}\delta_j^{-a}$). No doubt about the frozen I-G1 arose: it is proved as stated; ν is used only through $\hat{\mathfrak{c}}_*$.

4. Applying I-G1: local energy charts and uniform constants

This part turns the isoenergetic a-posteriori theorem (Theorem 1.14, proved in Part 3 as Theorem 3.17) into the family of invariant tori that the confinement proof of I-HEAD (Part 5) consumes. The data are those of the frozen I-HEAD (Theorem 1.16) with $n = 2$; $H = h + f$ is the perturbed Hamiltonian, and the approximate torus is the flat torus $K(\theta) = (\theta, I)$. Four things are proved.

(1) *Local energy charts* (§4.2). The nondegeneracy $\det \mathsf{B}_h \neq 0$ forces $\nabla h \neq 0$ and makes $I \mapsto (h(I), \text{direction of } \nabla h(I))$ a local diffeomorphism (Lemma 4.7). A finite cover by such charts (Lemma 4.9) is the only place where compactness of $\overline{B'}$ enters.
(2) *Two Diophantine directions* (§4.3). In each chart there are Diophantine vectors $\omega_\pm^*$ in two open sectors on either side of the direction of ∇h; this uses the measure theorem G3 of track K with the fixed exponent $\tau = 2 > n-1$ (finding I05 of the Part-0 review; no density is claimed at $\tau = 1$). For every energy E in the window of the chart the two points $I_\pm(E)$ on the curve $\{h = E\}$ with $\nabla h(I_\pm) \parallel \omega_\pm^*$ exist (Lemma 4.13); no implicit function theorem with a parameter is needed.
(3) *Verification of (I1)–(I13)* for $H = h + f$ and $K(\theta) = (\theta, I_\pm)$, $e = C\varepsilon$ (§4.4), using Part 2: $\mathsf{T}_H = \mathsf{B}_h + O(\varepsilon)$ (Lemmas 2.6, 2.8), and the Cauchy estimates and the Neumann series of track K.
(4) *Uniform constants* (§4.5). All constants of Theorem 1.14 are chosen once, per ray, before f, ε, E, I_0; ε_1 is the minimum over the finitely many rays and the other smallness conditions. The output is Proposition 4.19, whose conclusion (T1)–(T7) is what Part 5 needs.

Source: The local charts, the finite cover and the choice of constants are ours (ledger D-I-CONFINEMENT). Input from the sources: the isoenergetic nondegeneracy is HL19 (3.13)–(3.14) and Rem. 3.7, p. 19, and the bordered determinant of Chi05 p. 17 (Kepler case); both are about the extended torsion, not about charts.
Source gap: Neither HL19 nor Chi05 passes from $\det \mathsf{B}_h \neq 0$ to the existence of invariant tori on both sides of a given point of an energy surface; Lemmas 4.4–4.13 supply the bridge.

4.1. **Standing data, constants, external facts.**

Standing assumption 4.1 (Data of Part 4)**.** The data of Theorem 1.16 with $n = 2$: $B \subset \mathbb{R}^2$ open, bounded, nonempty; $s_0, r_0 > 0$; h holomorphic on B_{r_0} with $\operatorname{Im} h(I) = 0$ and $\det \mathsf{B}_h(I) \neq 0$ for every real $I \in B_{r_0}$; $B' \subset \mathbb{R}^2$ open with $\overline{B'}$ compact, $\overline{B'} \subset B$; $\eta > 0$. We assume $B' \neq \emptyset$ (otherwise the conclusion of I-HEAD is vacuous and no patch is needed). The exponent is fixed:

$$\tau := 2 \quad (> n - 1 = 1).$$

The perturbation is $f : \mathbb{C}^2 \times \mathbb{C}^2 \to \mathbb{C}$, $\varepsilon \in \mathbb{R}$ as in I-HEAD (holomorphic on $T_{s_0} \times B_{r_0}$, 1-periodic in θ there, real at real θ and real $I \in B_{r_0}$, $0 \leq \varepsilon$, $|f| \leq \varepsilon$ on $T_{s_0} \times B_{r_0}$), and $H := h + f$, i.e. $H(\theta, I) = h(I) + f(\theta, I)$. Write $\iota : \mathbb{R}^2 \to \mathbb{C}^2$ for the inclusion and $U := \iota^{-1}(B_{r_0}) \subset \mathbb{R}^2$, open, $B \subset U$ ([K:`bp:p4:freq-map`](i): $\iota B \subset B_{r_0}$, since $r_0 > 0$).

The setting of track K's Part 4 ([K:`bp:p4:data`]) is exactly this one (with $n = 2$, B open bounded, h holomorphic on B_{r_0}, f as above), apart from its Kolmogorov and boundary-layer hypotheses, which are not used here. We therefore take its shrunken domains and constants verbatim; they are listed so that the proofs below can quote them.

Definition 4.2 (Constants of the data)**.** *(a) From* [K:`bp:p4:data`]. $s := \min\{s_0, 1\}$, $\delta_* := \frac{1}{4}\min\{s, r_0\}$, and

$$\mu := \tfrac{1}{2}\delta_*, \qquad \rho := \tfrac{s}{4}, \qquad \rho_\infty := \tfrac{s}{8}, \qquad \mathfrak{D} := T_{s/2} \times B_{r_0/2}, \qquad \mathfrak{D}' := T_{3s/4} \times B_{3r_0/4}.$$

$M_h := \sup\{|h(I)| : I \in B_{3r_0/4}\} < \infty$ ([K:`bp:p4:h-bounded`]); $c(m, j)$ are the constants of [K:`bp:p1:cauchy-open`], $\bar{c} := \max\{1, c(2, j), c(4, j) : j = 1, 2, 3\}$, $F_j := \bar{c}\,\delta_*^{-j}$ $(j = 1, 2, 3)$, and $M := \bar{c}(M_h + 1)\delta_*^{-3}$. Note $\delta_* \leq \frac{1}{4}$, $\mu \leq \frac{1}{8}\min\{s, r_0\}$, $\rho_\infty < \rho$, $F_1 \leq F_2$.
(b) The compact action set. $d_B := \inf\{\|I - J\| : I \in \overline{B'}, J \in \mathbb{R}^2 \setminus B\}$ if $B \neq \mathbb{R}^2$ (always, since B is bounded); $d_B > 0$ because $\overline{B'}$ is compact and contained in the open set B. Put

$$\eta_1 := \min\{\eta,\ d_B/2\}, \qquad \mathcal{S} := \{I \in \mathbb{R}^2 : \operatorname{dist}(I, \overline{B'}) \leq d_B/2\}.$$

(c) Gradient and bordered inverse on $\mathcal{S}$.

$$m_h := \min_{I \in \mathcal{S}} \|\nabla h(I)\|, \qquad M_g := \max_{I \in \mathcal{S}} \|\nabla h(I)\|, \qquad S_h := \max_{I \in \mathcal{S}} |\mathsf{B}_h(I)^{-1}| \quad \text{(row-sum norm)}.$$

(d) Data for I-G1.

$$\nu := \tfrac{1}{4}m_h, \qquad \sigma_\omega := \tfrac{4M_g}{m_h} + 2, \qquad \sigma_K := 1+\mu, \qquad \sigma_B := 1+\mu, \qquad \sigma_T := 2S_h + \mu,$$

$$\varepsilon_T := \frac{1}{4S_hF_2}, \qquad e := F_1\,\varepsilon \quad \text{(the error bound of I-G1, } C_e := F_1\text{)}.$$

All of these depend only on $(B, s_0, r_0, h, B', \eta)$ (in fact η enters only η_1), and not on f, ε or any point.

Lemma 4.3 ($\mathcal{S}$ and the constants are well defined)**.** *$\mathcal{S}$ is compact, $\overline{B'} \subset \mathcal{S} \subset B$, and $0 < m_h \le M_g < \infty$, $0 < S_h < \infty$.*

Proof. **Step S4.1: Compactness and inclusion.** $I \mapsto \operatorname{dist}(I, \overline{B'})$ is continuous, so $\mathcal{S}$ is closed; it is bounded since $\overline{B'}$ is. If $I \notin B$ then $\|I - J\| \ge d_B$ for the nearest $J \in \overline{B'}$ (definition of d_B), so $\operatorname{dist}(I, \overline{B'}) \ge d_B > d_B/2$; hence $\mathcal{S} \subset B$.

Step S4.2: The gradient is nonzero. By Lemma 4.4(a) below (which does not use this lemma), $\nabla h(I) \neq 0$ for every $I \in U \supset \mathcal{S}$. ∇h is continuous on U ([K:`bp:p4:freq-map`](iii): it is differentiable), so $\|\nabla h\|$ attains a positive minimum and a finite maximum on the compact set $\mathcal{S}$.

Step S4.3: The bordered inverse. The entries of $\mathsf{B}_h(I)$ are continuous on U ([K:`bp:p4:freq-map`](iii) and the continuity of the second derivatives of the holomorphic h, [K:`bp:p1:holomorphic-regularity`]), and $\det \mathsf{B}_h \neq 0$ on U (hypothesis, $\iota U \subset B_{r_0}$). By Cramer's rule ([K:`bp:p1:ext-cramer`]), $\mathsf{B}_h^{-1} = \operatorname{adj}\mathsf{B}_h / \det \mathsf{B}_h$ has continuous entries on U, which are real ([K:`bp:p1:real-inverse`]). The row-sum norm is continuous, hence bounded on $\mathcal{S}$. □

External facts used in this part. In addition to [K:`bp:p1:ext-cramer`], [K:`bp:p1:ext-integral`] and the external facts of Part 2:

(Y1) *Inverse function theorem.* If $\Omega \subset \mathbb{R}^m$ is open, $F : \Omega \to \mathbb{R}^m$ is C^1 and $DF(c)$ is invertible, then there is an open $W \ni c$, $W \subset \Omega$, such that $F|_W$ is a homeomorphism onto an open set $F(W)$ with C^1 inverse. (Mathlib: `HasStrictFDerivAt.toOpenPartialHomeomorph`.) The restriction of such an $F|_W$ to an open $W' \subset W$ has the same property.

(Y2) *Compactness.* A continuous function on a compact set attains its extrema; a compact subset of $\mathbb{R}^2$ covered by open sets is covered by finitely many of them (Heine–Borel).

(Y3) *Lebesgue measure on $\mathbb{R}^2$.* It is monotone and countably subadditive on outer measures; a nonempty open set has positive measure; a bounded set has finite measure.

4.2. **Local energy charts.** The bordered Hessian is, in dimension two, the curvature of the energy curve times a positive factor; we use only its determinant. For $g = (g_1, g_2) \in \mathbb{R}^2$ put $g^\perp := (-g_2, g_1)$ and $\det(g, w) := g_1w_2 - g_2w_1 = g^\perp \cdot w$, where $\cdot$ is the Euclidean dot product on $\mathbb{R}^2$.

Lemma 4.4 (The bordered determinant)**.** *Let $I \in U$ and $g := \nabla h(I)$, $\nabla^2 h(I) := D(\nabla h)(I)$ (real symmetric,* [K:`bp:p4:freq-map`]*(iii)). Then*

$$\det \mathsf{B}_h(I) = -\,g^\perp \cdot \nabla^2 h(I)\, g^\perp.$$

Consequently (a) $\nabla h(I) \neq 0$ whenever $\det \mathsf{B}_h(I) \neq 0$ (in particular on all of U); (b) if $g \neq 0$, then $\det \mathsf{B}_h(I) \neq 0$ iff $\det\big(g, \nabla^2 h(I) g^\perp\big) \neq 0$, and $\det(g, \nabla^2 h(I) g^\perp) = g^\perp \cdot \nabla^2 h(I) g^\perp = -\det \mathsf{B}_h(I)$.

Proof. **Step S4.4: Expansion.** Write $h_a = \partial_a h(I)$, $h_{ab} = \partial_a\partial_b h(I)$ (real, $h_{12} = h_{21}$). By Definition 1.7 and Lemma 2.6 (Step S2.10), $\mathsf{B}_h = \begin{pmatrix} h_{11} & h_{12} & h_1 \\ h_{12} & h_{22} & h_2 \\ h_1 & h_2 & 0 \end{pmatrix}$. Expanding along the last row, $\det \mathsf{B}_h = h_1(h_{12}h_2 - h_1h_{22}) - h_2(h_{11}h_2 - h_1h_{12}) = -(h_{22}h_1^2 - 2h_{12}h_1h_2 + h_{11}h_2^2)$. With $g^\perp = (-h_2, h_1)$, $g^\perp \cdot \nabla^2 h\, g^\perp = h_{11}h_2^2 - 2h_{12}h_1h_2 + h_{22}h_1^2$.

Step S4.5: Consequences. (a) If $g = 0$ the right-hand side is 0. (b) is the identity $\det(g, w) = g^{\perp} \cdot w$ with $w = \nabla^2 h\, g^{\perp}$. □

Definition 4.5 (Direction coordinates at a point)**.** Let $c \in \overline{B'}$ and $g_c := \nabla h(c) \neq 0$, $q_c := g_c \cdot g_c > 0$. For $w \in \mathbb{R}^2$ put

$$a_c(w) := \frac{g_c \cdot w}{q_c}, \qquad b_c(w) := \frac{\det(g_c, w)}{q_c}, \qquad \text{so that } w = a_c(w)\, g_c + b_c(w)\, g_c^{\perp}.$$

(check: first component $\big(g_1(g\cdot w) - g_2 \det(g,w)\big)/q = w_1$, similarly the second). On $\{w : a_c(w) > 0\}$ define the *slope* $\sigma_c(w) := b_c(w)/a_c(w) = \det(g_c, w)/(g_c \cdot w)$, and on

$$\mathcal{U}_c := \{I \in U :\ a_c(\nabla h(I)) > 0\} \quad (\text{open},\ c \in \mathcal{U}_c,\ \text{since } a_c(g_c) = 1)$$

the *chart map*

$$m_c(I) := \sigma_c(\nabla h(I)), \qquad \Phi_c(I) := \big(h(I),\, m_c(I)\big) \in \mathbb{R}^2, \qquad I \in \mathcal{U}_c.$$

Since ∇h is C^1 on U ([K:bp:p4:freq-map]), Φ_c is C^1 on $\mathcal{U}_c$; $\Phi_c(c) = (h(c), 0)$.

Remark 4.6 (Slopes encode rays). If w, w' have $a_c(w), a_c(w') > 0$, then $\sigma_c(w) = \sigma_c(w')$ iff $w' = \lambda w$ for some $\lambda > 0$, and then $\lambda = a_c(w')/a_c(w)$ (from $w = a(g + \sigma g^{\perp})$). Hence for $I \in \mathcal{U}_c$ and a vector ω^* with $a_c(\omega^*) > 0$: $m_c(I) = \sigma_c(\omega^*)$ iff $\nabla h(I) = s\,\omega^*$ with $s = a_c(\nabla h(I))/a_c(\omega^*) > 0$, and then, for the max norm, $s = \|\nabla h(I)\|/\|\omega^*\|$.

Lemma 4.7 (Local chart)**.** *For every $c \in \overline{B'}$ the derivative $D\Phi_c(c)$ is invertible, with $\det D\Phi_c(c) = -\det \mathsf{B}_h(c)/q_c \neq 0$. Hence there is an open $W \subset \mathcal{U}_c$ with $c \in W$ such that $\Phi_c|_W : W \to V := \Phi_c(W)$ is a homeomorphism onto an open set with C^1 inverse, and W may be taken inside any prescribed open neighbourhood of c.*
Source: ours (the isoenergetic nondegeneracy as a chart); the determinant identity is the one of Chi05 p. 17 in the Kepler case, HL19 Rem. 3.7, p. 19.

Proof. **Step S4.6: Derivative of h and of m_c.** $Dh(c)u = g_c \cdot u$ ([K:bp:p4:freq-map](ii)). The slope map σ_c has numerator $\det(g_c, \cdot)$ vanishing at g_c and denominator $g_c \cdot w = q_c$ at $w = g_c$, so $D\sigma_c(g_c)[\delta w] = \det(g_c, \delta w)/q_c$. By the chain rule and $D(\nabla h)(c) = \nabla^2 h(c)$, $Dm_c(c)u = \det(g_c, \nabla^2 h(c)u)/q_c = g_c^{\perp} \cdot \nabla^2 h(c) u / q_c = (\nabla^2 h(c) g_c^{\perp}) \cdot u / q_c$ by the symmetry of $\nabla^2 h(c)$.

Step S4.7: Determinant. $D\Phi_c(c)$ has the rows $g_c^{\top}$ and $(\nabla^2 h(c) g_c^{\perp})^{\top}/q_c$, so $\det D\Phi_c(c) = \det(g_c, \nabla^2 h(c) g_c^{\perp})/q_c = -\det \mathsf{B}_h(c)/q_c$ by Lemma 4.4(b). It is nonzero because $c \in \overline{B'} \subset B \subset U$ and $\iota U \subset B_{r_0}$.

Step S4.8: Inverse function theorem. $\mathcal{U}_c$ is open, Φ_c is C^1 on it, and (Y1) gives W; intersecting with a prescribed open neighbourhood of c keeps the conclusion. □

Definition 4.8 (Patch)**.** A *patch* is a tuple $\mathfrak{p} = (c, W, \delta_E, \delta_m, \rho_c)$ with $c \in \overline{B'}$, W as in Lemma 4.7, $\delta_E, \delta_m, \rho_c > 0$, such that, with $V = \Phi_c(W)$,

- (P1) $W \subset \{I \in \mathbb{R}^2 : \|I - c\| < \eta_1/4\}$ (so $W \subset \mathcal{S}$);
- (P2) the box $Q := [h(c) - \delta_E,\, h(c) + \delta_E] \times [-\delta_m, \delta_m]$ lies in V;
- (P3) $\rho_c \le \eta_1/4$, the ball $\{\|I - c\| < \rho_c\}$ lies in W, and on it $|h(I) - h(c)| < \delta_E/2$ and $|m_c(I)| < \delta_m/4$.

We write $\Phi_{\mathfrak{p}}^{-1} : V \to W$ for the inverse.

Lemma 4.9 (Finite cover)**.** *There are finitely many patches $\mathfrak{p}_j = (c_j, W_j, \delta_{E,j}, \delta_{m,j}, \rho_j)$, $j \in \mathcal{P}$, with $\overline{B'} \subset \bigcup_j \{\|I - c_j\| < \rho_j\}$. The family depends only on $(B, s_0, r_0, h, B', \eta)$.*
Source: ours.

Proof. **Step S4.9: A patch at every $c \in \overline{B'}$.** Fix c. Lemma 4.7 with the neighbourhood $\{\|I-c\| < \eta_1/4\}$ gives W with (P1); V is open and contains $\Phi_c(c) = (h(c), 0)$, so (P2) holds for δ_E, δ_m small. W is open and h, m_c are continuous on $\mathcal{U}_c$ with $m_c(c) = 0$, so (P3) holds for ρ_c small.

Step S4.10: Compactness. The open balls $\{\|I - c\| < \rho_c\}$, $c \in \overline{B'}$, cover the compact set $\overline{B'}$; take a finite subcover ((Y2)). □

4.3. **Two Diophantine directions and the two points on each energy curve.**

Lemma 4.10 (Diophantine vectors in an open set)**.** *Let $\tau = 2$ and let $O \subset \mathbb{R}^2$ be nonempty, open and bounded. Then there are $\gamma > 0$ and $\omega \in O \cap \mathcal{D}_{\gamma,2}$.*
Source: standard; measure input is G3 of track K, Pos01 §1d, p. 5, via [K:bp:p0:g3]*. At the endpoint $\tau = n-1 = 1$ G3 is not available and this argument gives nothing (Part-0 review, I05): we do not use $\tau = 1$.*
Frozen Lean declaration: `KAM.volume_not_diophantine_le`

Proof. **Step S4.11: Positive measure.** $0 < \operatorname{meas} O < \infty$ ((Y3)). [K:bp:p0:g3] with $n = 2$, $\tau = 2 > 1$ and $\Omega = O$ gives $C \geq 0$ with $\operatorname{meas}\{\omega \in O : \omega \notin \mathcal{D}_{\alpha,2}\} \leq C\alpha$ for all $\alpha > 0$.

Step S4.12: Choice of γ. Take $\gamma := \operatorname{meas} O/(2(C+1))$. Then $\operatorname{meas}\{\omega \in O : \omega \notin \mathcal{D}_{\gamma,2}\} \leq C\gamma < \operatorname{meas} O$, so by subadditivity $O \cap \mathcal{D}_{\gamma,2} \neq \emptyset$. □

Lemma 4.11 (Rays in a sector)**.** *Let $c \in \overline{B'}$, $J \subset \mathbb{R}$ a nonempty open bounded interval. Then there are $\gamma > 0$ and $\omega^* \in \mathbb{R}^2$ with*

$$\omega^* \in \mathcal{D}_{\gamma,2}, \qquad a_c(\omega^*) > 0, \qquad \sigma_c(\omega^*) \in J, \qquad \tfrac{1}{4}m_h < \|\omega^*\| < \tfrac{1}{2}m_h.$$

Proof. **Step S4.13: An open nonempty set.** $O := \{\omega \in \mathbb{R}^2 : a_c(\omega) > 0,\ \sigma_c(\omega) \in J,\ \frac{1}{4}m_h < \|\omega\| < \frac{1}{2}m_h\}$ is open (preimage of open sets under continuous maps on the open set $\{a_c > 0\}$) and bounded. For $\varsigma \in J$ and $t > 0$ put $\omega_t = t(g_c + \varsigma g_c^{\perp})$. Then $a_c(\omega_t) = t$, $\sigma_c(\omega_t) = \varsigma$, and $\|\omega_t\| = t\|g_c + \varsigma g_c^{\perp}\|$ with $\|g_c + \varsigma g_c^{\perp}\| > 0$ (its Euclidean length is $\sqrt{q_c(1+\varsigma^2)}$). Choosing $t = \frac{3}{8}m_h/\|g_c + \varsigma g_c^{\perp}\|$ gives $\omega_t \in O$.

Step S4.14: Diophantine. Lemma 4.10 applied to O. □

Definition 4.12 (Rays of a patch)**.** For each patch $\mathfrak{p} = (c, W, \delta_E, \delta_m, \rho_c)$ of Lemma 4.9 choose, by Lemma 4.11 with $J_+ = (\delta_m/2, \delta_m)$ and $J_- = (-\delta_m, -\delta_m/2)$, vectors $\omega^*_{\mathfrak{p},\pm}$ and constants $\gamma_{\mathfrak{p},\pm} > 0$ with $\omega^*_{\mathfrak{p},\pm} \in \mathcal{D}_{\gamma_{\mathfrak{p},\pm},2}$, $a_c(\omega^*_{\mathfrak{p},\pm}) > 0$, $\frac{1}{4}m_h < \|\omega^*_{\mathfrak{p},\pm}\| < \frac{1}{2}m_h$, and put

$$m_{\mathfrak{p},\pm} := \sigma_c(\omega^*_{\mathfrak{p},\pm}), \qquad \text{so } m_{\mathfrak{p},+} \in (\tfrac{\delta_m}{2}, \delta_m), \quad m_{\mathfrak{p},-} \in (-\delta_m, -\tfrac{\delta_m}{2}).$$

There are finitely many rays ($2|\mathcal{P}|$); the choice is made once and depends only on $(B, s_0, r_0, h, B', \eta)$.

Lemma 4.13 (The two points on the energy curve)**.** *Let $\mathfrak{p}$ be a patch, $E \in \mathbb{R}$ with $|E - h(c)| \leq \delta_E$, and write $\omega^*_{\pm}, m_{\pm}$ for the data of Definition 4.12. Then $(E, m_{\pm}) \in Q \subset V$, so*

$$I_{\pm}(E) := \Phi_{\mathfrak{p}}^{-1}(E, m_{\pm}) \in W \quad (\textit{real points of } \mathbb{R}^2)$$

are defined, and

- *(a) $h(I_{\pm}) = E$, $m_c(I_{\pm}) = m_{\pm}$, and $\nabla h(I_{\pm}) = s^{\circ}_{\pm}\,\omega^*_{\pm}$ with $s^{\circ}_{\pm} = \|\nabla h(I_{\pm})\|/\|\omega^*_{\pm}\| \in \big(2,\ 4M_g/m_h\big)$;*
- *(b) $\|I_{\pm} - c\| < \eta_1/4$, $I_{\pm} \in \mathcal{S} \subset B$, and $I_- \neq I_+$; for every I_0 with $\|I_0 - c\| < \rho_c$: $\|I_{\pm} - I_0\| < \eta_1/2$;*
- *(c) (ordering) $m_c(I_-) \leq -\delta_m/2$, $m_c(I_+) \geq \delta_m/2$, whereas $|m_c(I_0)| < \delta_m/4$ for $\|I_0 - c\| < \rho_c$.*

Moreover $\Phi_{\mathfrak{p}}$ maps $\{I \in W :\ h(I) = E,\ |m_c(I)| \leq \delta_m\}$ homeomorphically onto $\{E\} \times [-\delta_m, \delta_m]$, so that in the chart the energy curve is the segment $m \mapsto (E, m)$ and I_-, I_+ are the points with $m = m_{\pm}$ ("I_-, I_0, I_+ in this order along the curve").
Source: ours.

Proof. **Step S4.15: Existence.** $|m_\pm| < \delta_m$ and $|E - h(c)| \le \delta_E$, so $(E, m_\pm) \in Q \subset V = \Phi_c(W)$ by (P2).

Step S4.16: Parallel gradient. $\Phi_c(I_\pm) = (E, m_\pm)$ gives $h(I_\pm) = E$ and $m_c(I_\pm) = \sigma_c(\omega_\pm^*)$, with $a_c(\nabla h(I_\pm)) > 0$ since $I_\pm \in W \subset \mathcal{U}_c$ and $a_c(\omega_\pm^*) > 0$. Remark 4.6 gives $\nabla h(I_\pm) = s_\pm^\circ \omega_\pm^*$ with $s_\pm^\circ = \|\nabla h(I_\pm)\|/\|\omega_\pm^*\|$. By (P1), $I_\pm \in \mathcal{S}$ (since $\eta_1/4 \le d_B/8$), so $m_h \le \|\nabla h(I_\pm)\| \le M_g$, while $\frac{1}{4} m_h < \|\omega_\pm^*\| < \frac{1}{2} m_h$; hence $2 < s_\pm^\circ < 4M_g/m_h$.

Step S4.17: Position. $I_\pm \in W$ and (P1) give $\|I_\pm - c\| < \eta_1/4$; with $\|I_0 - c\| < \rho_c \le \eta_1/4$ ((P3)), $\|I_\pm - I_0\| < \eta_1/2$. $I_- \ne I_+$ since $m_c(I_-) < 0 < m_c(I_+)$. Ordering: (c) is $m_\pm$ vs. (P3).

Step S4.18: Energy segment. $\Phi_{\mathfrak{p}} : W \to V$ is a homeomorphism, and $\{E\} \times [-\delta_m, \delta_m] \subset Q \subset V$, whose preimage is exactly $\{I \in W : h(I) = E, |m_c(I)| \le \delta_m\}$. □

4.4. **The approximate torus and the hypotheses of I-G1.**

Lemma 4.14 (The hypotheses (I1)–(I4) for $H = h + f$)**.** *Let Assumption 4.1 hold with $\varepsilon \le 1$ and $D := \mathfrak{D} = T_{s/2} \times B_{r_0/2}$. Then (I1)–(I4) of Theorem 1.14 hold for (D, H) with the constant M of Definition 4.2. Moreover for $z \in \mathfrak{D}$ and $j = 1, 2, 3$: $\|D^j f(z)\| \le F_j \varepsilon$, in particular $\|X_f(z)\| \le F_1 \varepsilon$ and $\|D^2 f(z)\| \le F_2 \varepsilon$.*
Source: ours, quoting [K:bp:p4:h1] *and* [K:bp:p4:derivative-bounds].

Proof. **Step S4.19: (I1)–(I3).** Steps [K:step:p4:h1-open] and [K:step:p4:h1-holo] of [K:bp:p4:h1]: D is open and invariant under $\theta \mapsto \theta + k$; H is holomorphic on $D \subset T_{s_0} \times B_{r_0}$, 1-periodic in θ, and real at real phase points of D (for real θ and real $I \in B_{r_0}$: $h(I)$ and $f(\theta, I)$ are real by the hypotheses of I-HEAD).

Step S4.20: (I4). By [K:bp:p4:derivative-bounds](a), (c), (d) (Step [K:step:p4:db-cauchy]) with $\varepsilon \le 1$: $\|DH\|, \|X_H\|, \|DX_H\|, \|D^2 X_H\| \le M$ on D. The same step gives $\|D^j H\| \le \bar{c}(M_h + \varepsilon)\delta_*^{-j} \le M$ ($\delta_* \le 1$) for $j = 2$, which is the bound on $\|D^2 H\|$ in (I4). The bounds $\|D^j f\| \le F_j \varepsilon$ are [K:bp:p4:derivative-bounds](c). □

Lemma 4.15 (The flat torus of $H = h + f$)**.** *Let Assumption 4.1 hold with $\varepsilon \le 1$, $D = \mathfrak{D}$. Let $I \in B$ (a real point), $\omega^* \in \mathbb{R}^2$, $s' \in \mathbb{R}$ with $\nabla h(I) = s'\omega^*$, $\omega := s'\omega^*$, and $K(\theta) := (\theta, I)$ for $\theta \in \mathbb{C}^2$. Then for $\theta \in T_{s/2}$:*

(a) K is holomorphic, primary and real on reals; $K(T_{s/2}) \subset D$; $DK(\theta)v = (v, 0)$, $\|DK(\theta)\| = 1$; $G_K \equiv I_2$, $|G_K^{-1}| = 1$; $N_{\cdot j} \equiv (0, e_j)$. For $\theta \in T_\rho$: $B_\mu(K(\theta)) \subset D$.
(b) $E_{H,K,\omega}(\theta) = X_f(K(\theta)) = \big((\partial_{I_a} f)_a, (-\partial_{\theta_a} f)_a\big)(\theta, I)$ and $\|E_{H,K,\omega}(\theta)\| \le F_1 \varepsilon$.
(c) $\mathsf{T}_H(\theta) = \mathsf{B}_h(I) + R(\theta)$ with $R(\theta) = \begin{pmatrix} (\partial_{I_i} \partial_{I_j} f)_{ij} & 0 \\ (\partial_{I_b} f)_b^\top & 0 \end{pmatrix}(\theta, I)$, $|R(\theta)| \le 2F_2\varepsilon$, hence $|\langle \mathsf{T}_H \rangle - \mathsf{B}_h(I)| \le 2F_2\varepsilon$.
(d) If $I \in \mathcal{S}$ and $\varepsilon \le \varepsilon_T$, then $\det\langle \mathsf{T}_H \rangle \ne 0$ and $|\langle \mathsf{T}_H \rangle^{-1}| \le 2S_h = \sigma_T - \mu$.
(e) $|\bar{E}_H(K) - h(I)| \le \varepsilon$; hence $|\bar{E}_H(K) - E_0| \le \varepsilon$ whenever $h(I) = E_0$.

Source: ours; Part 2, Lemmas 2.6 and 2.8 for the frame and the last row; HL19 Rem. 3.7, p. 19.

Proof. **Step S4.21: The torus and its frame.** Steps S2.5 and S2.6 of Lemma 2.6 apply verbatim (K and the frame do not involve the Hamiltonian): K is affine, primary, real on reals (I real), $DK = (\mathrm{id}, 0)$, $G_K = I_2$, $N_{\cdot j} = (0, e_j)$ constant. $\|DK(\theta)v\| = \|(v, 0)\| = \|v\|$. For $\theta \in T_{s/2}$ and $I \in B$, $K(\theta) \in T_{s/2} \times B \subset D$. If $\theta \in T_\rho$ and $\|(\theta', I') - K(\theta)\| < \mu$, then $|\mathrm{Im}\, \theta_i'| < s/4 + \mu < s/2$ ($\mu \le s/8$) and $\|I' - I\| < \mu \le r_0/8$, so $I' \in B_{r_0/2}$: $B_\mu(K(\theta)) \subset D$.

Step S4.22: The error. $X_H(\theta, I) = \big((\partial_{I_a} h + \partial_{I_a} f)_a, (-\partial_{\theta_a} f)_a\big)$ and $(\partial_{I_a} h(I))_a = \nabla h(I) = \omega$ (Lemma 2.6, Step S2.7); with $DK\omega = (\omega, 0)$ we get $E = X_f(K(\theta))$, and $\|X_f\| \le F_1 \varepsilon$ on D (Lemma 4.14).

Step S4.23: Torsion and border. $DX_H(z)(0,e_j) = \big((\partial_{I_a}\partial_{I_j}H)_a, (-\partial_{\theta_a}\partial_{I_j}H)_a\big)$, $\Omega_0(x,y) = (-y,x)$, $D_\theta N_{\cdot j} = 0$. So $T_{ij} = (0,e_i)\cdot\Omega_0 DX_H N_{\cdot j} = \partial_{I_i}\partial_{I_j}H = \partial_i\partial_j h(I) + \partial_{I_i}\partial_{I_j}f(\theta,I)$ (as in Step S2.8). The last column is $\omega = \nabla h(I)$. By Lemma 2.8(a), the last row is $Y = \omega^\top - (\eta^L)^\top$ with $\eta^L_b = -N_{\cdot b}^\top \Omega_0 E = -\partial_{I_b} f$; i.e. $Y_b = \partial_b h(I) + \partial_{I_b} f(\theta,I)$ (equal to the direct value $DH(K)N_{\cdot b}$). The upper-left block of $\mathsf{B}_h(I)$ is $(\partial_a\partial_b h(I))$ and its border is $\nabla h(I)$ (Step S2.10), which gives (c)'s identity. For the size: row $i \le 2$ of R has absolute sum $\sum_j |D^2 f[e_i,e_j]| \le 2\|D^2 f\| \le 2F_2\varepsilon$, the last row $\sum_b |\partial_{I_b} f| \le 2F_1\varepsilon \le 2F_2\varepsilon$ (Lemma 4.14). The average over $[0,1]^2 \subset \mathbb{R}^2 \subset T_{s/2}$ is integrable (Lemma 2.5(a)) and the norm of an average is at most the supremum ([K:bp:p1:ext-integral]); $\langle \mathsf{B}_h \rangle = \mathsf{B}_h$ since the cube has volume one.

Step S4.24: Invertibility. For $I \in \mathcal{S}$, $|\mathsf{B}_h(I)^{-1}| \le S_h$ and $|\mathsf{B}_h^{-1}||\langle R\rangle| \le S_h \cdot 2F_2\varepsilon \le \frac{1}{2}$ when $\varepsilon \le \varepsilon_T$. [K:bp:p1:neumann] gives invertibility of $\mathsf{B}_h(I) + \langle R\rangle = \langle \mathsf{T}_H\rangle$ and $|\langle \mathsf{T}_H\rangle^{-1}| \le 2S_h$ (a real matrix, [K:bp:p1:real-inverse]); $2S_h = \sigma_T - \mu$.

Step S4.25: Energy average. $H(K(\theta)) = h(I) + f(\theta,I)$ and for $\theta \in [0,1]^2$, $(\theta,I) \in T_{s_0}\times B_{r_0}$ so $|f(\theta,I)| \le \varepsilon$; hence $|\bar{E}_H(K) - h(I)| = |\int f(\theta,I)\,d\theta| \le \varepsilon$. □

Lemma 4.16 (One application of I-G1)**.** *Let $\mathfrak{p}$ be a patch with ray $\omega^* = \omega^*_{\mathfrak{p},\pm}$, $\gamma = \gamma_{\mathfrak{p},\pm}$. Apply Theorem 1.14 with $n=2$, $\tau=2$, $\rho = s/4$, $\rho_\infty = s/8$ and the constants $M, \sigma_K, \sigma_B, \sigma_T, \mu, \sigma_\omega, \nu$ of Definition 4.2, $\bar\gamma := \gamma$, ω^* as given. Its constants $C_1, C_2, C_3 > 0$ (denoted $C_i^{\mathfrak{p},\pm}$) depend only on $(B, s_0, r_0, h, B', \eta)$ and the ray. Then, for every f, ε of Assumption 4.1 with*

$$\varepsilon \le 1, \qquad \varepsilon \le \varepsilon_T, \qquad \varepsilon < \varepsilon^{\mathfrak{p},\pm} := \frac{\gamma^4\rho^8}{C_1^{\mathfrak{p},\pm}F_1},$$

every $E \in \mathbb{R}$ with $|E - h(c)| \le \delta_E$, and $I := I_\pm(E)$, $s' := s^\circ_\pm$ of Lemma 4.13, the hypotheses (I1)–(I13) of Theorem 1.14 hold for

$$D = \mathfrak{D}, \quad H = h + f, \quad K(\theta) = (\theta, I), \quad s = s^\circ_\pm, \quad \gamma, \quad e = F_1\varepsilon, \quad E_0 = E.$$

Consequently there are K_∞ and $s_\infty \in (1, \sigma_\omega)$ with (O1)–(O6), in particular $\|K_\infty - K\| \le C_2F_1\varepsilon/(\gamma^2\rho^4)$ on T_{ρ_∞}, $\|(s_\infty - s^\circ_\pm)\omega^*\| \le C_3F_1\varepsilon/(\gamma^2\rho^4)$.

Proof. **Step S4.26: (I1)–(I4).** Lemma 4.14 ($\varepsilon \le 1$).

Step S4.27: (I5)–(I7). Lemma 4.15(a) with $I \in B$ (Lemma 4.13(b)): K is holomorphic, primary, real on reals on T_ρ; $B_\mu(K(\theta)) \subset D$ and $\|DK\| = 1 = \sigma_K - \mu$; $\det G_K = 1$, $|G_K^{-1}| = 1 = \sigma_B - \mu$.

Step S4.28: (I8). $\nabla h(I) = s^\circ_\pm\omega^*$ (Lemma 4.13(a)), $I \in \mathcal{S}$; Lemma 4.15(d) with $\omega = s^\circ_\pm\omega^*$ gives $\det\langle \mathsf{T}_H\rangle \neq 0$, $|\langle \mathsf{T}_H\rangle^{-1}| \le \sigma_T - \mu$ ($\varepsilon \le \varepsilon_T$).

Step S4.29: (I9). $1 < 2 < s < 4M_g/m_h < \sigma_\omega$. $(s-1)\|\omega^*\| > \|\omega^*\| > \frac{1}{4}m_h = \nu$ and $(\sigma_\omega - s)\|\omega^*\| > 2\|\omega^*\| > \nu$.

Step S4.30: (I10)–(I12). $0 < \gamma \le \bar\gamma = \gamma$, $\omega^* \in \mathcal{D}_{\gamma,2}$, $e = F_1\varepsilon \ge 0$ (Definition 4.12). (I11) is Lemma 4.15(b). (I12): $h(I) = E = E_0$, Lemma 4.15(e), and $\varepsilon \le F_1\varepsilon = e$ since $F_1 \ge 1$.

Step S4.31: (I13). $C_1 e/(\gamma^4\rho^{4\tau}) = C_1F_1\varepsilon/(\gamma^4\rho^8) < 1$ by $\varepsilon < \varepsilon^{\mathfrak{p},\pm}$ (and trivially if $\varepsilon = 0$).

Step S4.32: Conclusion. Theorem 1.14, with $C_2 e/(\gamma^2\rho^{2\tau}) = C_2F_1\varepsilon/(\gamma^2\rho^4)$. □

Remark 4.17 (Constants of Part 3 make the order of quantifiers harmless). The frozen Theorem 1.14 chooses C_1, C_2, C_3 after ω^* and $\bar\gamma$, which is why the application is per ray: finitely many rays, finitely many applications, ε_1 below is a minimum over a finite set. Theorem 3.17 shows that its constants do not depend on ω^* and σ_ω (only on $\bar\gamma$ through C_1); this is not needed.

4.5. **Uniform application: the output of Part 4.**

Definition 4.18 (Uniform constants)**.** With the patches $\mathfrak{p}_j$ ($j \in \mathcal{P}$) of Lemma 4.9, their rays (Definition 4.12) and the constants $C_i^{j,\pm}$ of Lemma 4.16, put

$$C_K := \max_{j\in\mathcal{P},\pm} \frac{F_1 \max\{C_2^{j,\pm}, C_3^{j,\pm}\}}{(\gamma_{j,\pm})^2\rho^4},$$

$$\varepsilon_1 := \min\Big\{1,\ \varepsilon_T,\ \min_j \tfrac{1}{2}\delta_{E,j},\ \min_{j,\pm} \varepsilon^{j,\pm},\ \frac{\eta_1}{2C_K}\Big\} \in (0,1].$$

Order of choice: $(B, s_0, r_0, h, B', \eta) \to \mathfrak{D}, \mu, \delta_*, M, F_j, \mathcal{S}, m_h, M_g, S_h, \nu, \sigma$'s $\to$ patches $\to$ rays, $\gamma_{j,\pm} \to C_i^{j,\pm} \to C_K, \varepsilon_1$. Neither depends on f, ε, E, I_0.

Proposition 4.19 (Invariant tori on both sides: output of Part 4)**.** *Let Assumption 4.1 hold (the data of I-HEAD with $n = 2$, $B' \neq \emptyset$, $\eta > 0$, $\tau = 2$). Let ε_1, C_K be as in Definition 4.18, $\rho_\infty = s/8$, $\mathcal{P}$, $\mathfrak{p}_j$ as above. Then for every $f : \mathbb{C}^2 \times \mathbb{C}^2 \to \mathbb{C}$ and ε with the hypotheses of I-HEAD and $\varepsilon < \varepsilon_1$, every $I_0 \in \overline{B'}$ and every $E \in \mathbb{R}$ with $|E - h(I_0)| \le \varepsilon$ there are $j \in \mathcal{P}$ with $\|I_0 - c_j\| < \rho_j$ and, for each sign $\pm$, a map $K_\pm : \mathbb{C}^2 \to \mathbb{C}^2 \times \mathbb{C}^2$, numbers $s_\pm \in (1, \sigma_\omega)$ and a real point $I_\pm \in \mathbb{R}^2$, such that, with $H = h + f$, $\omega_\pm^* = \omega_{j,\pm}^*$ and $\omega_\pm := s_\pm \omega_\pm^*$:*

- *(T1)* (torus) *$K_\pm$ is holomorphic on T_{ρ_∞}, primary on T_{ρ_∞} and real on reals; its restriction to $\mathbb{R}^2$ is real-analytic with values in $\mathbb{R}^2 \times \mathbb{R}^2$.*
- *(T2)* (invariance, Diophantine frequency) *$E_{H,K_\pm,\omega_\pm}(\theta) = 0$ for all $\theta \in T_{\rho_\infty}$; $\omega_\pm^* \in \mathcal{D}_{\gamma_{j,\pm},2}$, so $\omega_\pm$ is nonresonant; $\omega_\pm \in \mathbb{R}^2 \setminus \{0\}$.*
- *(T3)* (energy) *$H(K_\pm(\theta)) = E$ for all $\theta \in T_{\rho_\infty}$ (pointwise); in particular $\mathcal{H}_{h,f}(K_\pm(\theta)) = E$ for real θ.*
- *(T4)* (closeness to the flat torus; graph inputs) *$\|K_\pm(\theta) - (\theta, I_\pm)\| \le C_K\varepsilon$ on T_{ρ_∞}, and, writing $K_\pm(\theta) = (\theta + u_\pm(\theta), I_\pm + v_\pm(\theta))$ with $u_\pm, v_\pm : T_{\rho_\infty} \to \mathbb{C}^2$ holomorphic and 1-periodic, $\|u_\pm\|_{\rho_\infty}, \|v_\pm\|_{\rho_\infty} \le C_K\varepsilon$ and $\|Du_\pm\|_{\rho_\infty/2}, \|Dv_\pm\|_{\rho_\infty/2} \le 4C_K\varepsilon/\rho_\infty$ (operator norms; $\rho_\infty = s/8$).*
- *(T5)* (the base points and the frequencies) *$h(I_\pm) = E$, $\nabla h(I_\pm) = s_\pm^\circ \omega_\pm^*$ with $s_\pm^\circ \in (2, 4M_g/m_h)$, $\|\omega_\pm - \nabla h(I_\pm)\| = |s_\pm - s_\pm^\circ|\|\omega_\pm^*\| \le C_K\varepsilon$; and $\nabla h(I_\pm)$, $\omega_\pm$ are positive multiples of $\omega_\pm^*$.*
- *(T6)* (position) *$I_\pm \in B$, $\|I_\pm - I_0\| < \eta_1/2 \le \eta/2$, and for every real θ: $\|K_\pm(\theta)_2 - I_0\| < \eta$ and $K_\pm(\theta)_2 \in \mathcal{S} \subset B$ (the action component stays in B and within η of I_0). Moreover $B_{\mu/2}(K_\pm(\theta)) \subset \mathfrak{D}$ and $\|DK_\pm(\theta)\| \le \sigma_K - \mu/2$ on T_{ρ_∞}.*
- *(T7)* (ordering in the energy chart) *With the chart $\Phi_j = (h, m_{c_j})$ of $\mathfrak{p}_j$: $I_\pm = \Phi_j^{-1}(E, m_{j,\pm})$, $m_{c_j}(I_-) \le -\delta_{m,j}/2$, $m_{c_j}(I_+) \ge \delta_{m,j}/2$ and $|m_{c_j}(I_0)| < \delta_{m,j}/4$, so that, on the arc $\Phi_j^{-1}(\{E\} \times [-\delta_{m,j}, \delta_{m,j}])$ of the energy curve $\{h = E\}$ (the segment $m \mapsto (E, m)$ in the chart), the slope coordinate of I_- is below $-\delta_{m,j}/2$ and that of I_+ above $\delta_{m,j}/2$, while I_0 has slope coordinate in $(-\delta_{m,j}/4, \delta_{m,j}/4)$.*

The numbers ε_1, C_K, ρ_∞, $\mathcal{P}$, σ_ω, M_g depend only on $(B, s_0, r_0, h, B', \eta)$.
Source: ours (D-I-CONFINEMENT); the isoenergetic theorem used is HL19 Thm 3.6 and Rem. 3.7, pp. 18–19, in the form of Theorem 1.14.

Proof. **Step S4.33: Choice of the patch.** Since $I_0 \in \overline{B'}$ and the balls cover (Lemma 4.9), there is j with $\|I_0 - c_j\| < \rho_j$. Write $c = c_j$, $\delta_E = \delta_{E,j}$ etc. By (P3), $|h(I_0) - h(c)| < \delta_E/2$, hence $|E - h(c)| \le \varepsilon + \delta_E/2 \le \delta_E$ because $\varepsilon < \varepsilon_1 \le \delta_E/2$.

Step S4.34: The base points. Lemma 4.13 gives $I_\pm := I_\pm(E)$, $s_\pm^\circ$: this is (T5) (first part), (T7) and $I_\pm \in B$, $\|I_\pm - I_0\| < \eta_1/2$.

Step S4.35: Application of I-G1. $\varepsilon < \varepsilon_1 \le \min\{1, \varepsilon_T, \varepsilon^{j,\pm}\}$, so Lemma 4.16 gives $K_\pm := K_\infty$ and $s_\pm := s_\infty \in (1, \sigma_\omega)$ with (O1)–(O6) of Theorem 1.14 for $e = F_1\varepsilon$, $E_0 = E$. (T1): (O1), and the

real-analyticity by [K:bp:p1:real-restriction]. (T2): (O2); $\omega_\pm^* \in \mathcal{D}_{\gamma,2}$ is a property of the ray. (T3): (O3), and $\mathcal{H}_{h,f} = \operatorname{Re} H = H$ at real phase points of D whose value is real (Definition 1.4). (T5): (O6) with $C_3F_1\varepsilon/(\gamma^2\rho^4) \le C_K\varepsilon$; the multiples are positive since $s_\pm > 1$ and $s_\pm^\circ > 2$.

Step S4.36: Closeness and derivatives. (O5): $\|K_\pm - K\| \le C_2F_1\varepsilon/(\gamma^2\rho^4) \le C_K\varepsilon$ on T_{ρ_∞}, $K = (\theta, I_\pm)$. $u_\pm, v_\pm$ are the components of $K_\pm - K$, holomorphic on T_{ρ_∞}, 1-periodic because both $K_\pm$ and K are primary: $K_\pm(\theta + k) - K(\theta + k) = K_\pm(\theta) + (k, 0) - K(\theta) - (k, 0)$. The max norm gives $\|u\|, \|v\| \le C_K\varepsilon$. [K:bp:p1:cauchy-strip] with $n = 2$ and $\delta = \rho_\infty/2$ gives $\|Du\|_{\rho_\infty/2}, \|Dv\|_{\rho_\infty/2} \le 2\cdot\frac{2}{\rho_\infty}C_K\varepsilon = 4C_K\varepsilon/\rho_\infty$ (applied to the pair (u, v) as a vector-valued function bounded by $C_K\varepsilon$).

Step S4.37: Position. For real θ, $\|K_\pm(\theta)_2 - I_\pm\| \le C_K\varepsilon < \eta_1/2$ since $\varepsilon < \eta_1/(2C_K)$. With $\|I_\pm - I_0\| < \eta_1/2$ this gives $\|K_\pm(\theta)_2 - I_0\| < \eta_1 \le \eta$. Since $I_0 \in \overline{B'}$, $\operatorname{dist}(K_\pm(\theta)_2, \overline{B'}) < \eta_1 \le d_B/2$, so $K_\pm(\theta)_2 \in \mathcal{S} \subset B$ (Lemma 4.3). The remaining claims of (T6) are (O4) of Theorem 1.14 and the shape of σ_K: $\sigma_K - \mu/2$ and the ball $B_{\mu/2}$ are those of (O4), with $D = \mathfrak{D}$. □

Remark 4.20 (What Part 5 does with this). Given $z_0 = (\theta_0, I_0)$ with $I_0 \in B'$, its energy is $E := \mathcal{H}_{h,f}(z_0) = h(I_0) + \operatorname{Re} f(\theta_0, I_0)$, so $|E - h(I_0)| \le \varepsilon$ and Proposition 4.19 applies; take $\varepsilon_0 \le \varepsilon_1$. The two tori are invariant under the flow of $\mathcal{H}_{h,f}$ ($E_{H,K,\omega} = 0$ and $\mathcal{H} = H$ at real points), lie on the energy surface $\{\mathcal{H}_{h,f} = E\}$ of z_0, are graphs over $\mathbb{T}^2$ for ε small by (T4) (Part 5, p5.2), and sit on the two sides $m_c \le -\delta_m/2$, $m_c \ge \delta_m/2$ of z_0 in the energy chart (T7) (Part 5, p5.1, p5.3); the confinement $\|z(t)_2 - I_0\| < \eta$ is (T6) and the compact enclosure in $\mathcal{S} \subset B$ gives global existence. No positive measure of tori is used: only the existence of the $2|\mathcal{P}|$ Diophantine rays.

Remark 4.21 (Deviations, doubts). (1) $\tau = 2$, not $\tau \ge 1$ (review I05); (2) the hypothesis of I-HEAD that $\det \mathsf{B}_h \ne 0$ on all of the real slice of B_{r_0} is used only on $\mathcal{S} \subset B$ (and for $\nabla h \ne 0$, Lemma 4.4); (3) $\mathcal{H}_{h,f}$ is the real part of H, and $\mathcal{H}_{h,f} = H$ because H is real at real points of $\mathfrak{D}$ (I3); (4) the finite cover is not a Lebesgue-number argument: each patch has its own $\varepsilon^{j,\pm}$ and ε_1 is the minimum; (5) no frozen statement looks doubtful for this part. The only matrix identity we use beyond Part 2 is Lemma 4.4; the nonsingular-derivative argument of track N ([N:bpN:p2:qc-kernel]) is not needed.